

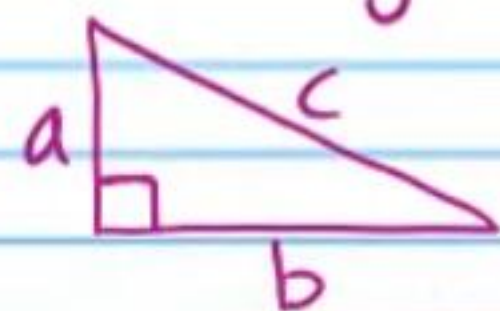
Trigonometry Review

Date
Course

Pythagorean Theorem

Sides $\longleftrightarrow$ Sides

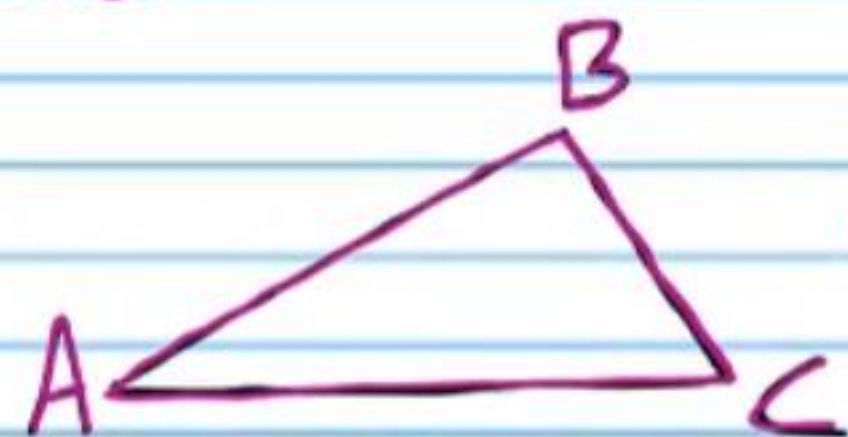
The Pythagorean Theorem: If a , b , and c are the legs and hypotenuse of a right triangle, respectively, then $a^2 + b^2 = c^2$.



Triangle Sum Theorem

Angles $\longleftrightarrow$ Angles

The Triangle Sum Theorem: The sum of the measures of the interior angles of any triangle is 180° .



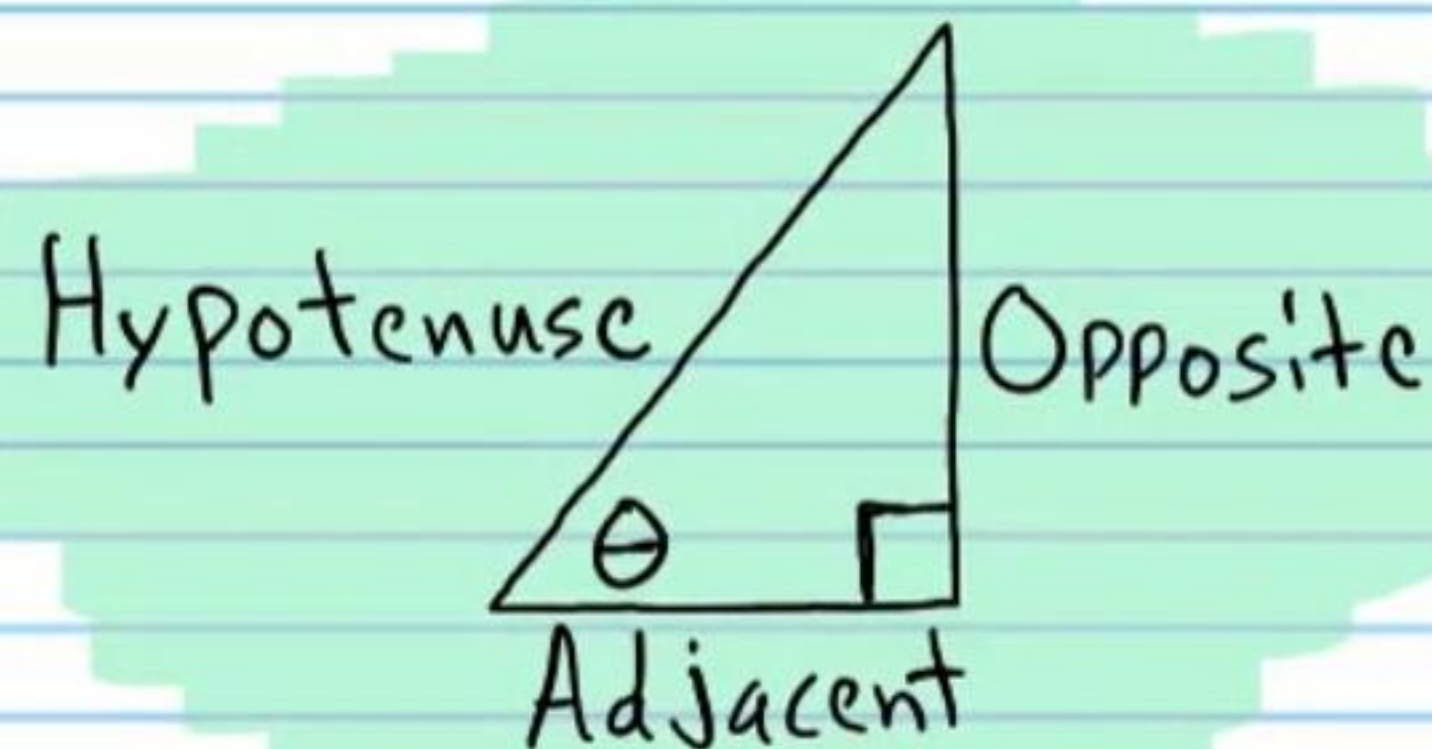
$$m\angle A + m\angle B + m\angle C = 180^\circ$$

Trigonometry

Sides $\longleftrightarrow$ Angles

Trigonometry: The study of relationships between sides and angles of triangles.

The Six Basic Trigonometric Functions



$$\sin \theta = \frac{\text{OPP}}{\text{HYP}}$$

$$\csc \theta = \frac{\text{HYP}}{\text{OPP}}$$

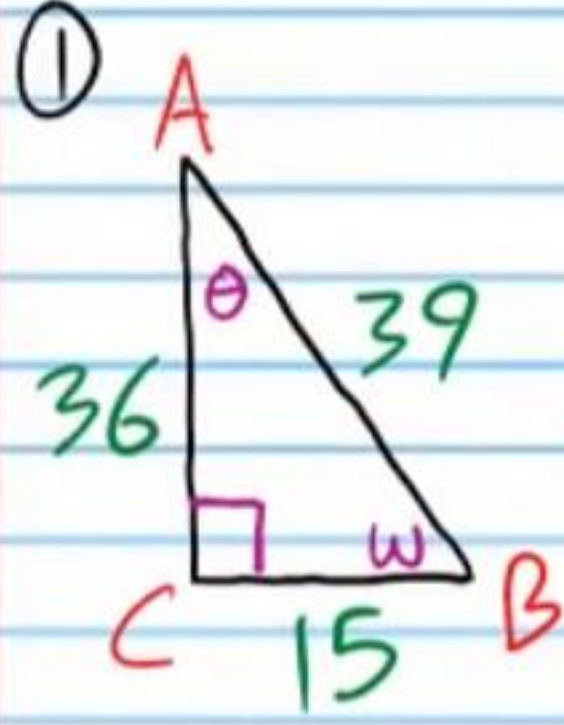
$$\cos \theta = \frac{\text{Adj}}{\text{HYP}}$$

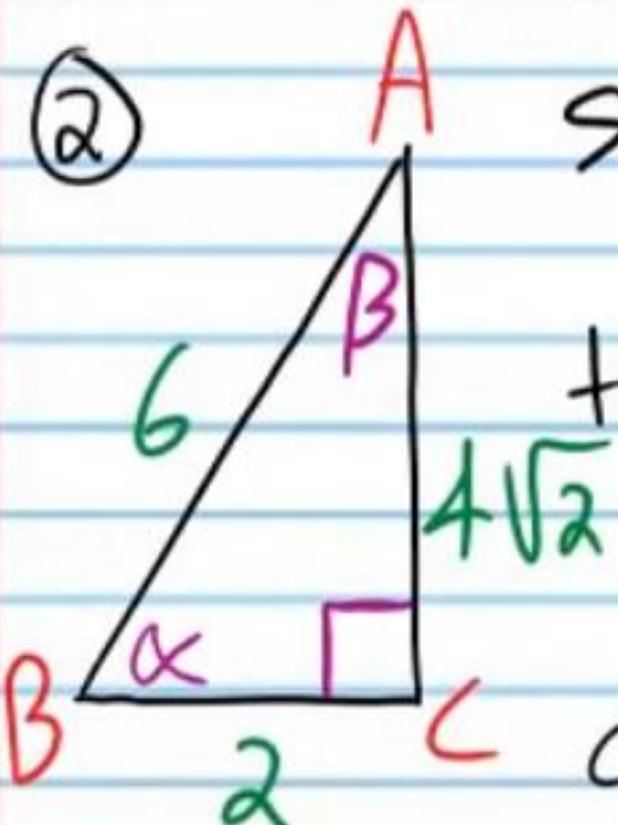
$$\sec \theta = \frac{\text{HYP}}{\text{Adj}}$$

$$\tan \theta = \frac{\text{OPP}}{\text{Adj}}$$

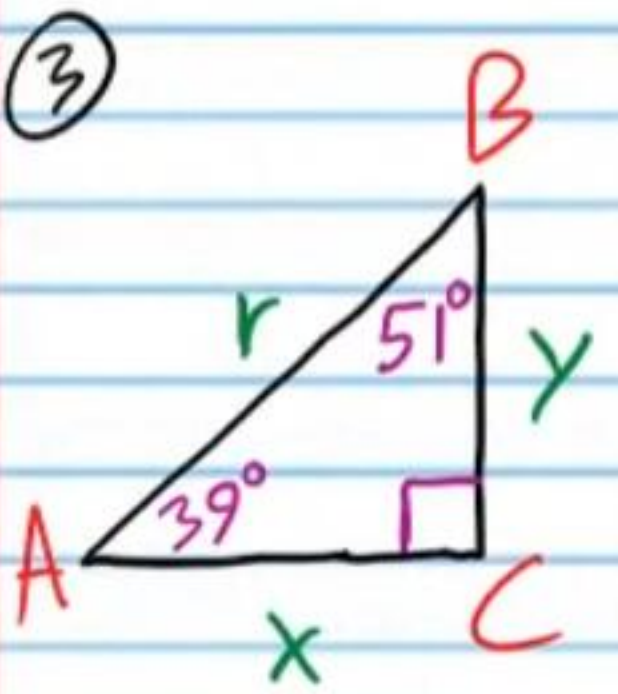
$$\cot \theta = \frac{\text{Adj}}{\text{OPP}}$$

Find the values of the expressions below.

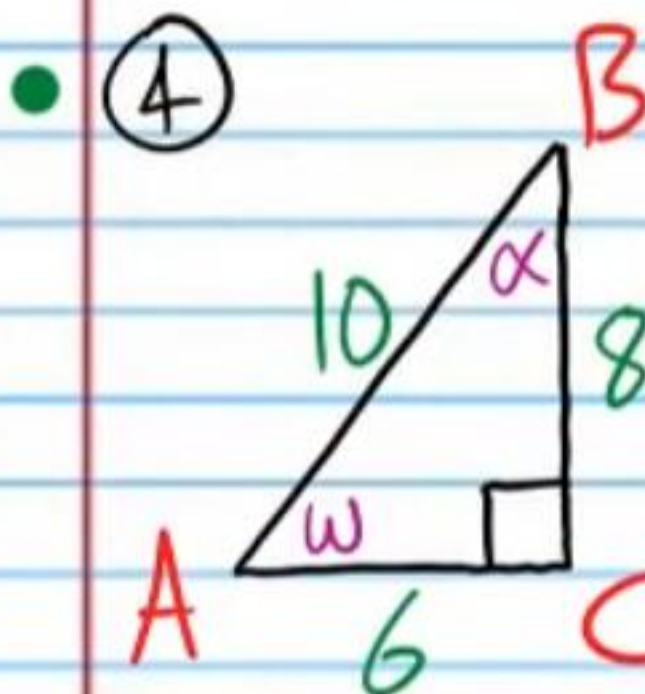
①  $\sin A = \frac{15}{39} = \boxed{\frac{5}{13}}$ $\cos \theta = \frac{36}{39} = \boxed{\frac{12}{13}}$
 $\tan B = \frac{36}{15} = \boxed{\frac{12}{5}}$ $\csc W = \frac{39}{36} = \boxed{\frac{13}{12}}$
 $\sec A = \frac{39}{15} = \boxed{\frac{13}{5}}$ $\cot \theta = \frac{26}{15} = \boxed{\frac{12}{5}}$

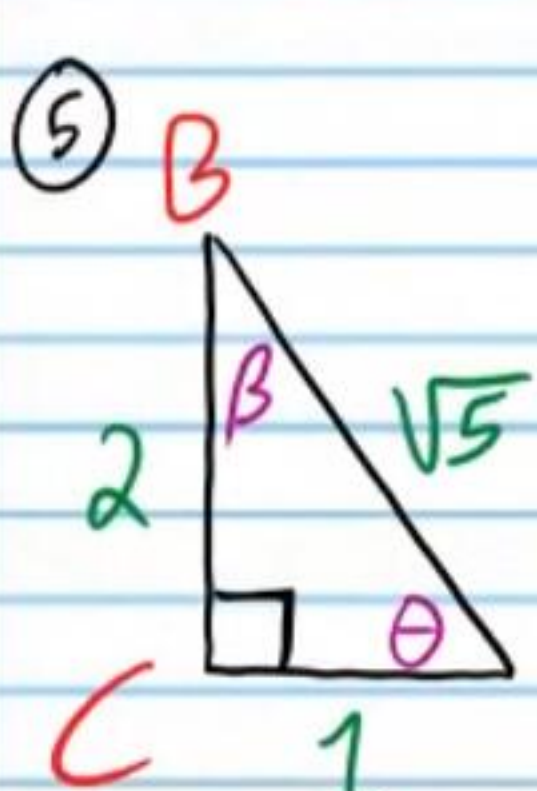
②  $\sin \alpha = \frac{4\sqrt{2}}{6} = \boxed{\frac{2\sqrt{2}}{3}}$ $\cos B = \frac{2}{6} = \boxed{\frac{1}{3}}$
 $\tan \beta = \frac{2}{4\sqrt{2}} = \frac{1}{2\sqrt{2}} = \frac{1}{2\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2\sqrt{4}} = \frac{\sqrt{2}}{2 \cdot 2} = \boxed{\frac{\sqrt{2}}{4}}$
 $\csc A = \frac{6}{2} = \boxed{3}$ $\sec \alpha = \frac{6}{2} = \boxed{3}$
 $\cot \beta = \frac{4\sqrt{2}}{2} = \boxed{2\sqrt{2}}$

Find an exact ratio for each expression listed below.

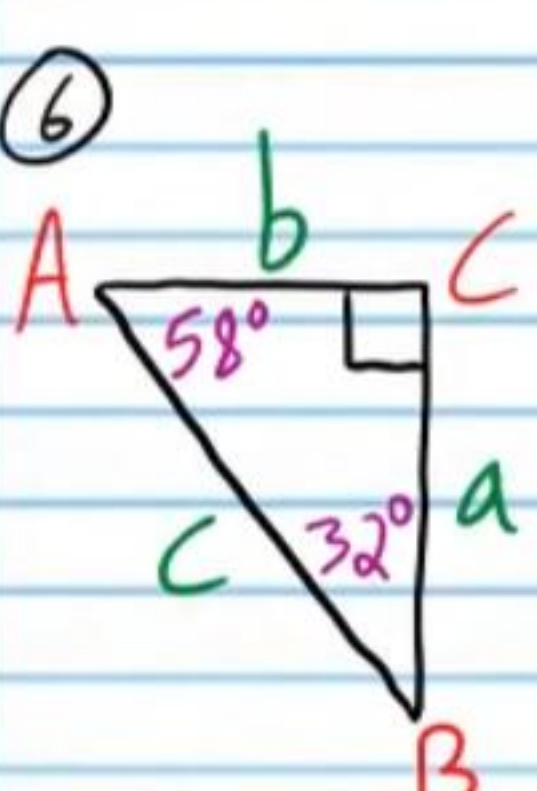
③  $\sin 39^\circ = \boxed{\frac{y}{r}}$ $\cos 39^\circ = \boxed{\frac{x}{r}}$
 $\tan A = \boxed{\frac{y}{x}}$ $\csc 51^\circ = \boxed{\frac{r}{y}}$
 $\sec B = \boxed{\frac{r}{x}}$ $\cot 51^\circ = \boxed{\frac{y}{x}}$

Find the values of the expressions below.

④  $\sin B = \frac{6}{10} = \boxed{\frac{3}{5}}$ $\cos \alpha = \frac{8}{10} = \boxed{\frac{4}{5}}$
 $\tan A = \frac{8}{6} = \boxed{\frac{4}{3}}$ $\csc \alpha = \frac{10}{6} = \boxed{\frac{5}{3}}$
 $\sec W = \frac{10}{6} = \boxed{\frac{5}{3}}$ $\cot B = \frac{8}{6} = \boxed{\frac{4}{3}}$

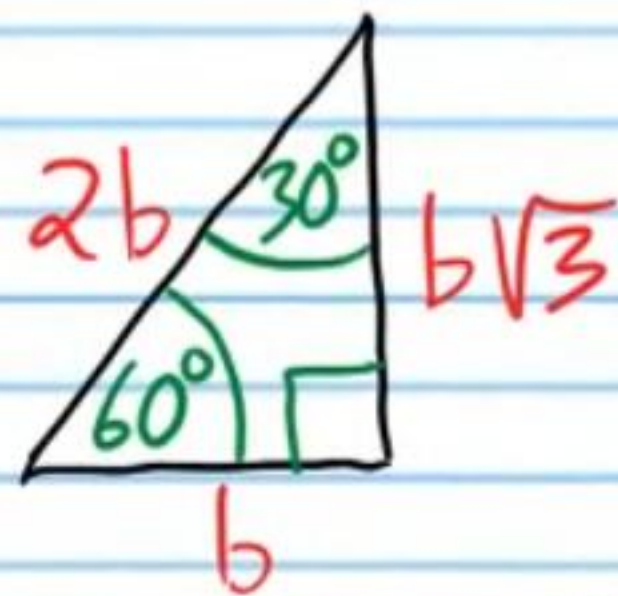
⑤  $\sin \beta = \frac{1}{\sqrt{5}} = \frac{1}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{5}}{\sqrt{25}} = \boxed{\frac{\sqrt{5}}{5}}$
 $\cos A = \frac{1}{\sqrt{5}} = \boxed{\frac{\sqrt{5}}{5}}$ $\tan B = \boxed{\frac{1}{2}}$
 $\csc \theta = \boxed{\frac{\sqrt{5}}{2}}$ $\sec B = \boxed{\frac{\sqrt{5}}{2}}$
 $\cot \beta = \frac{2}{1} = \boxed{2}$

Find an exact ratio for each expression listed below.

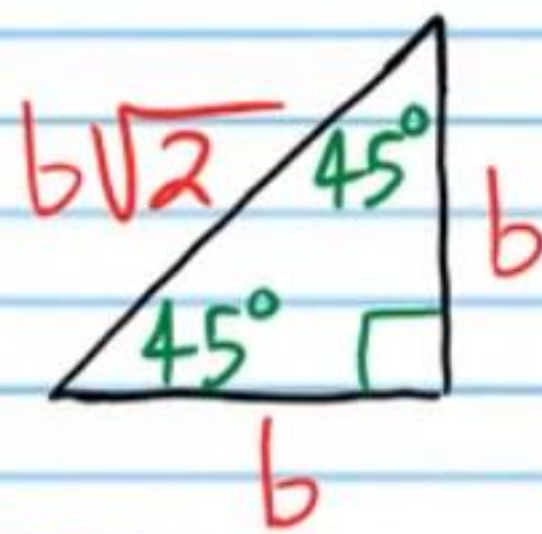
⑥  $\sin 58^\circ = \boxed{\frac{a}{c}}$ $\cos B = \boxed{\frac{a}{c}}$
 $\tan 32^\circ = \boxed{\frac{b}{a}}$ $\csc 32^\circ = \boxed{\frac{c}{b}}$
 $\sec A = \boxed{\frac{c}{b}}$ $\cot 58^\circ = \boxed{\frac{b}{a}}$

Special Right Triangles

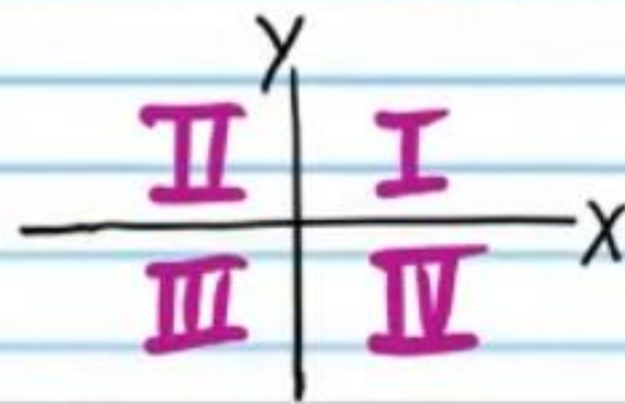
30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60°, and 90°, and the length of the side opposite the 30° angle is w , then the length of the hypotenuse is $2w$ and the length of the side opposite the 60° angle is $w\sqrt{3}$.



45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45°, and 90°, and the length of the two sides opposite the 45° angles are w , then the length of the hypotenuse is $w\sqrt{2}$.



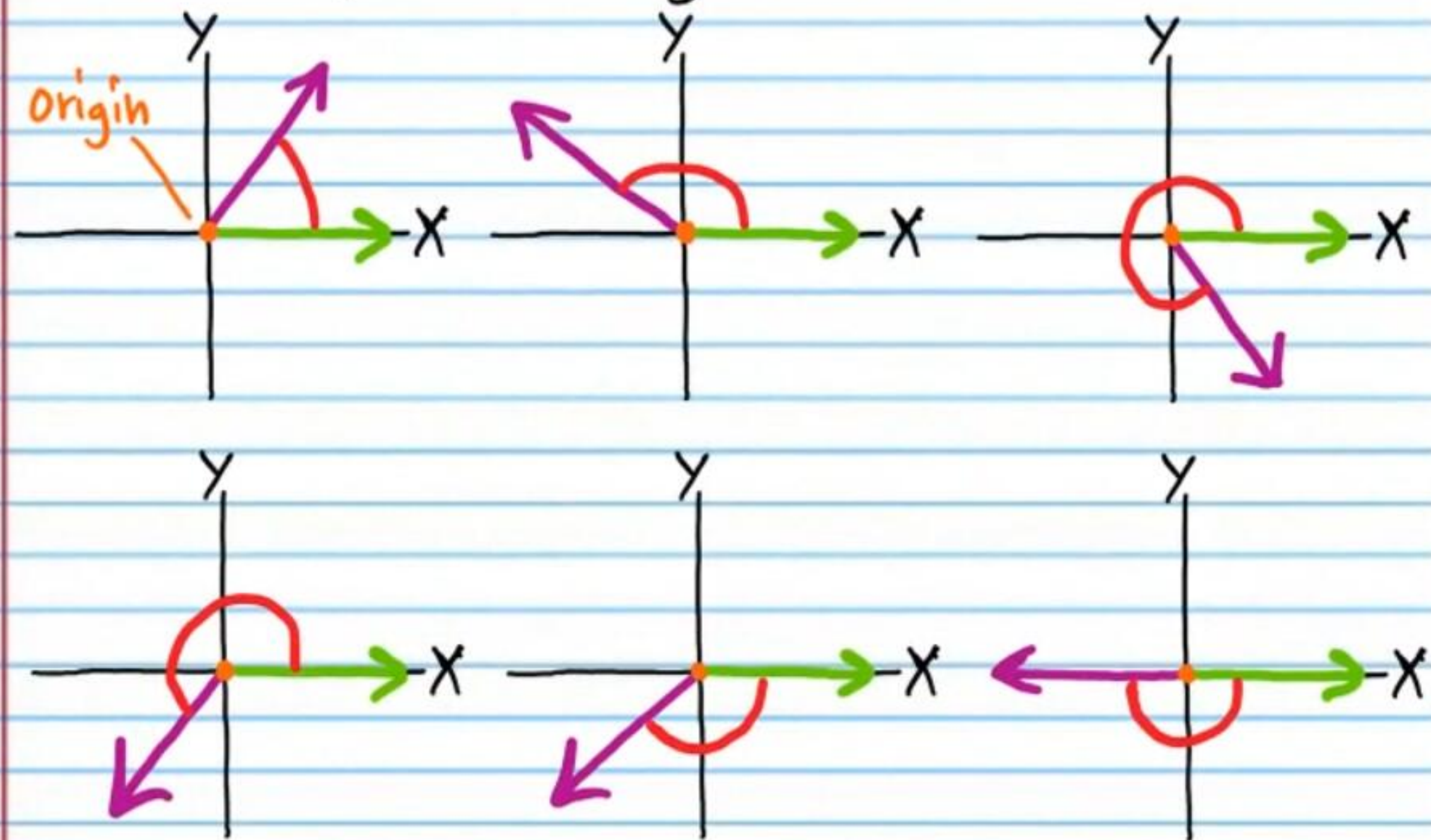
Quadrants of the XY Coordinate System



Angles in Standard Position

Standard Position of an Angle: Where an angle is placed so that its vertex is at the origin (i.e. $(0,0)$) of an XY coordinate system and its initial side is on the positive part of the X-axis.

Examples of Angles in Standard Position



Initial Side: The side of the angle that is placed on the positive part of the X-axis.

Terminal Side: The side of the angle where the angle measure stops (i.e. terminates).

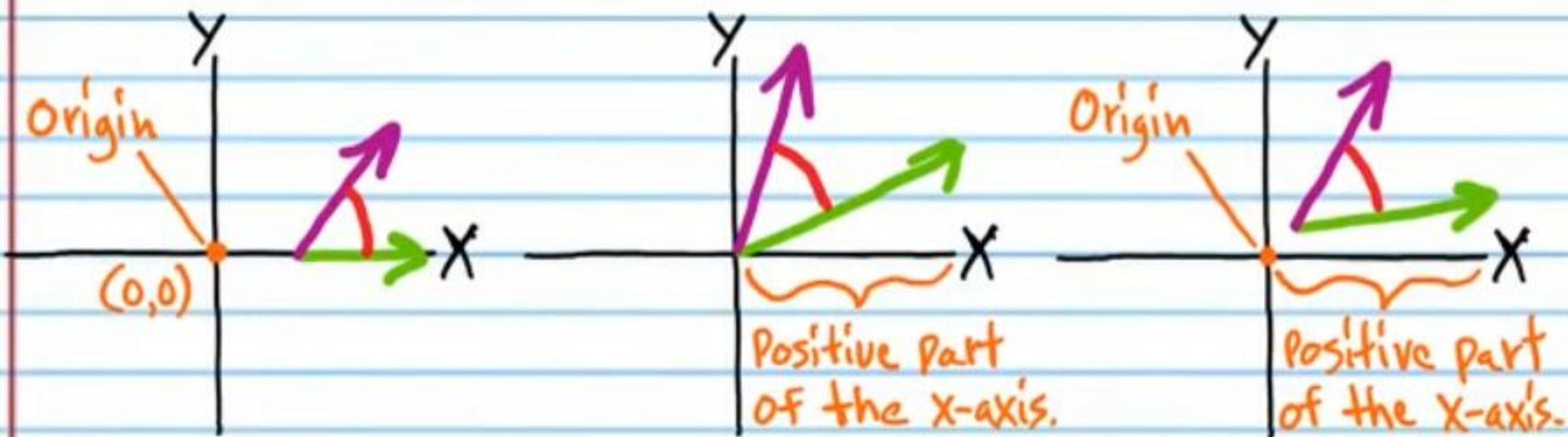
Angle Measure: The number of degrees (or radians) indicating the extent of rotation.

Examples of Angles that Are Not in Standard Position

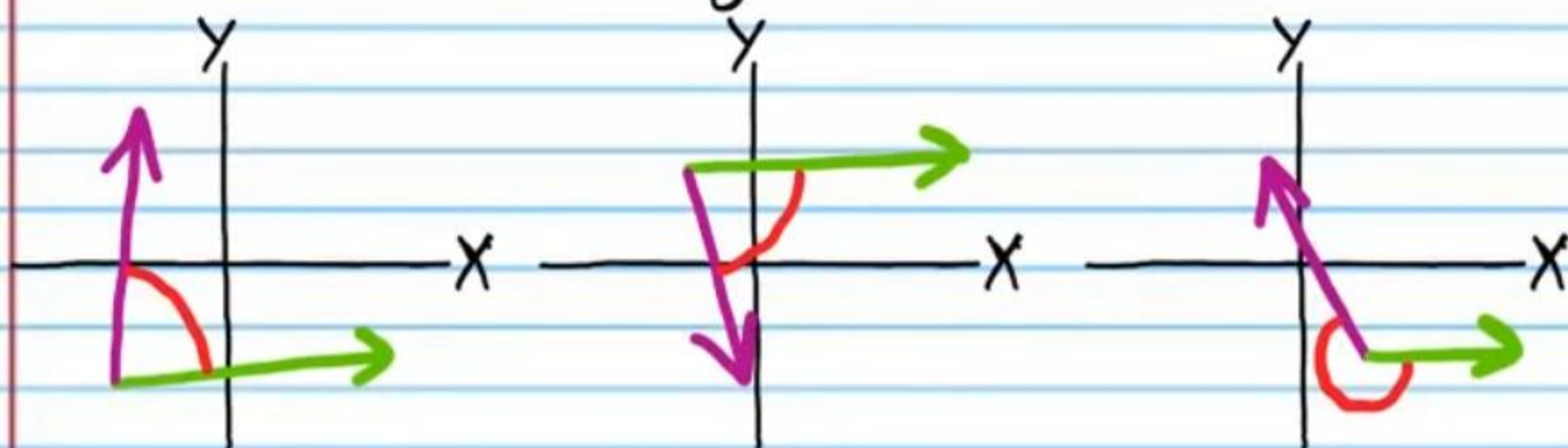
The vertex is not at the origin.

The initial side of the angle is not on the positive part of the x-axis.

The vertex is not at the origin and the initial side is not on the positive part of the x-axis.

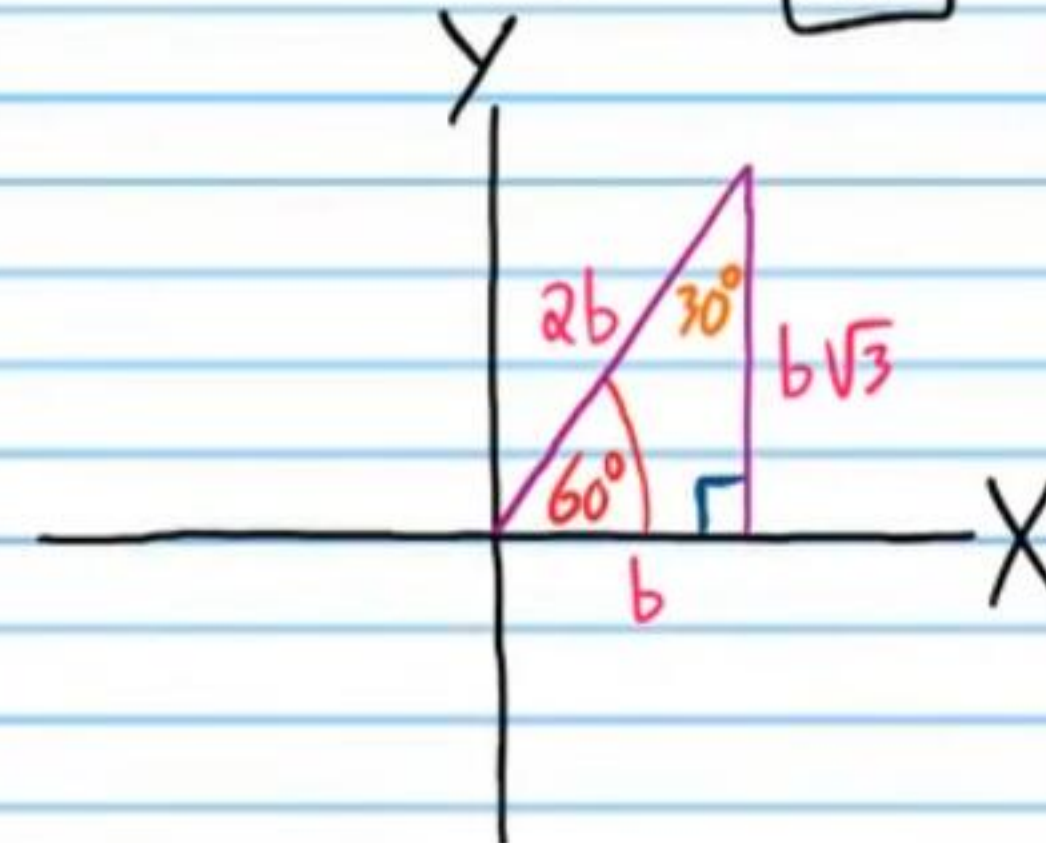


Other Examples of Angles Not in Standard Position

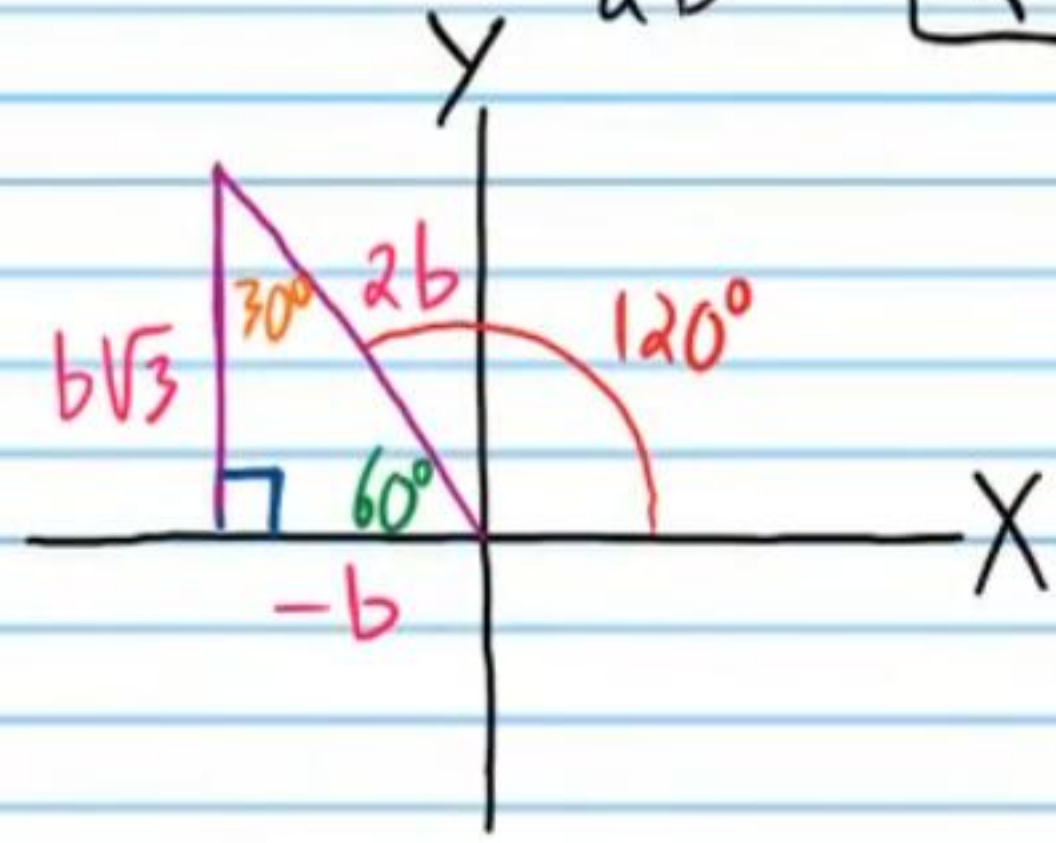


Examples

$$\sin 60^\circ = \frac{b\sqrt{3}}{2b} = \boxed{\frac{\sqrt{3}}{2}}$$

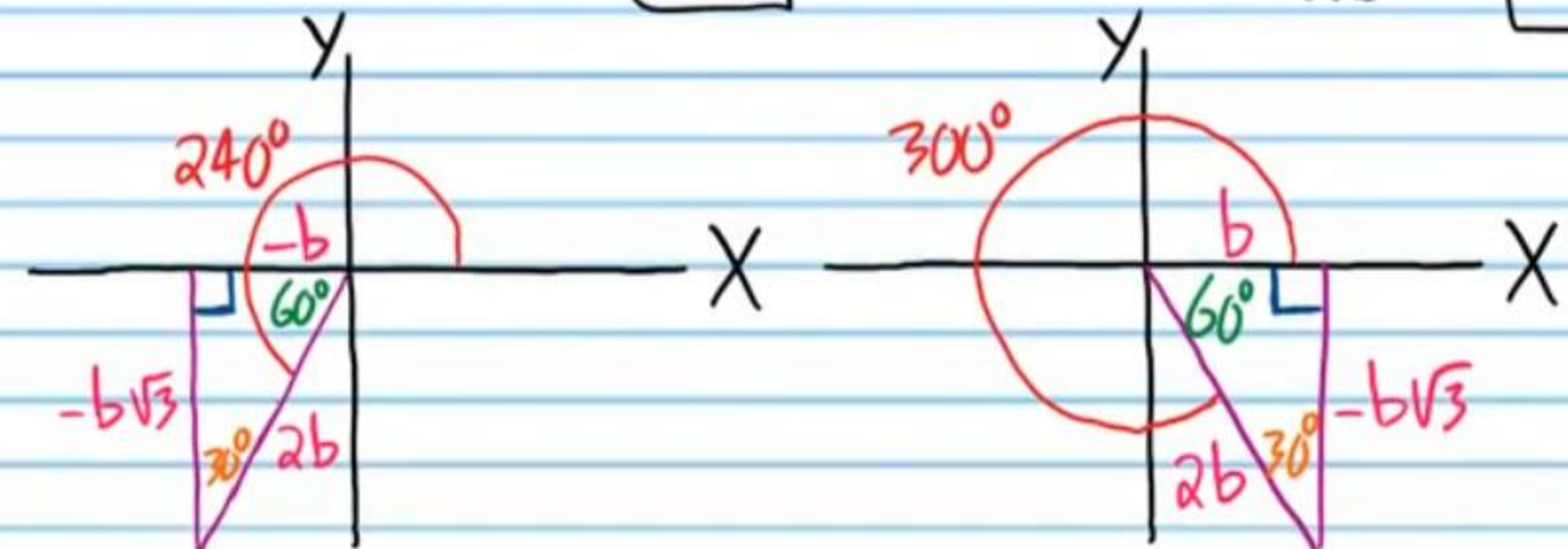


$$\sin 120^\circ = \frac{b\sqrt{3}}{2b} = \boxed{\frac{\sqrt{3}}{2}}$$



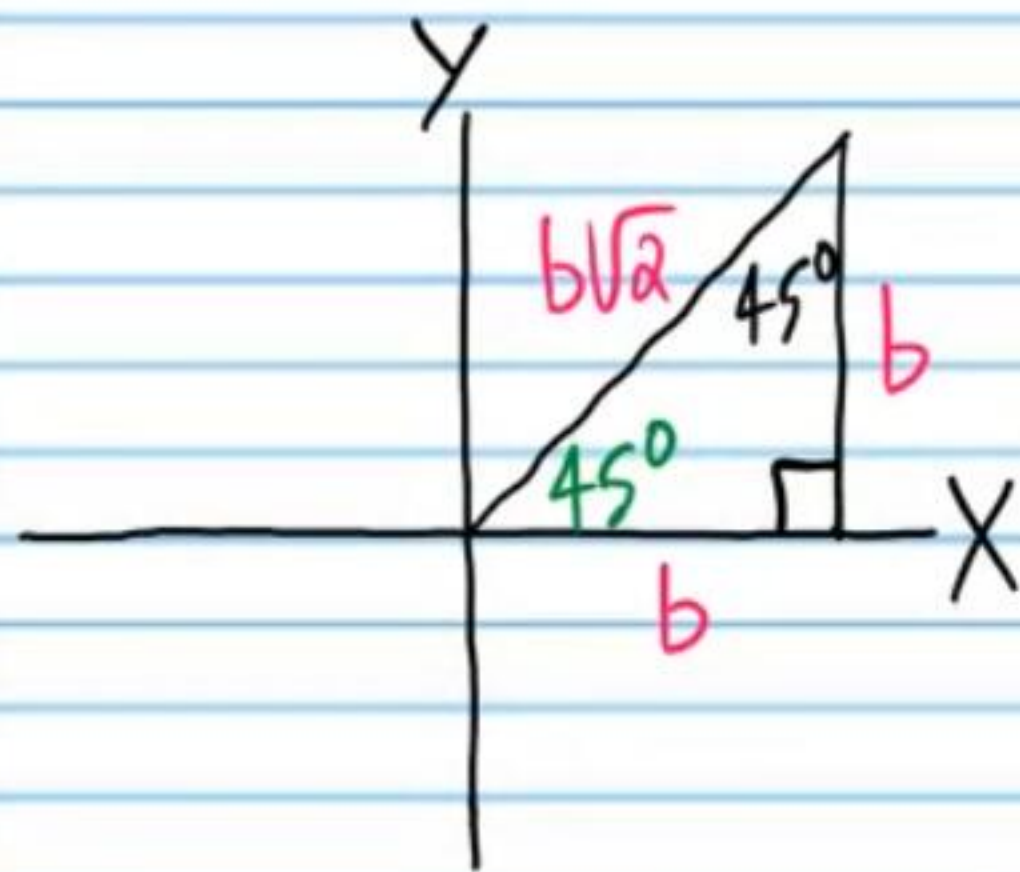
The Reference Angle of an Angle in Standard Position: The acute angle between the x-axis and the terminal side.

$$\sin 240^\circ = \frac{-b\sqrt{3}}{2b} = \boxed{-\frac{\sqrt{3}}{2}} \quad \sin 300^\circ = \frac{-b\sqrt{3}}{2b} = \boxed{-\frac{\sqrt{3}}{2}}$$

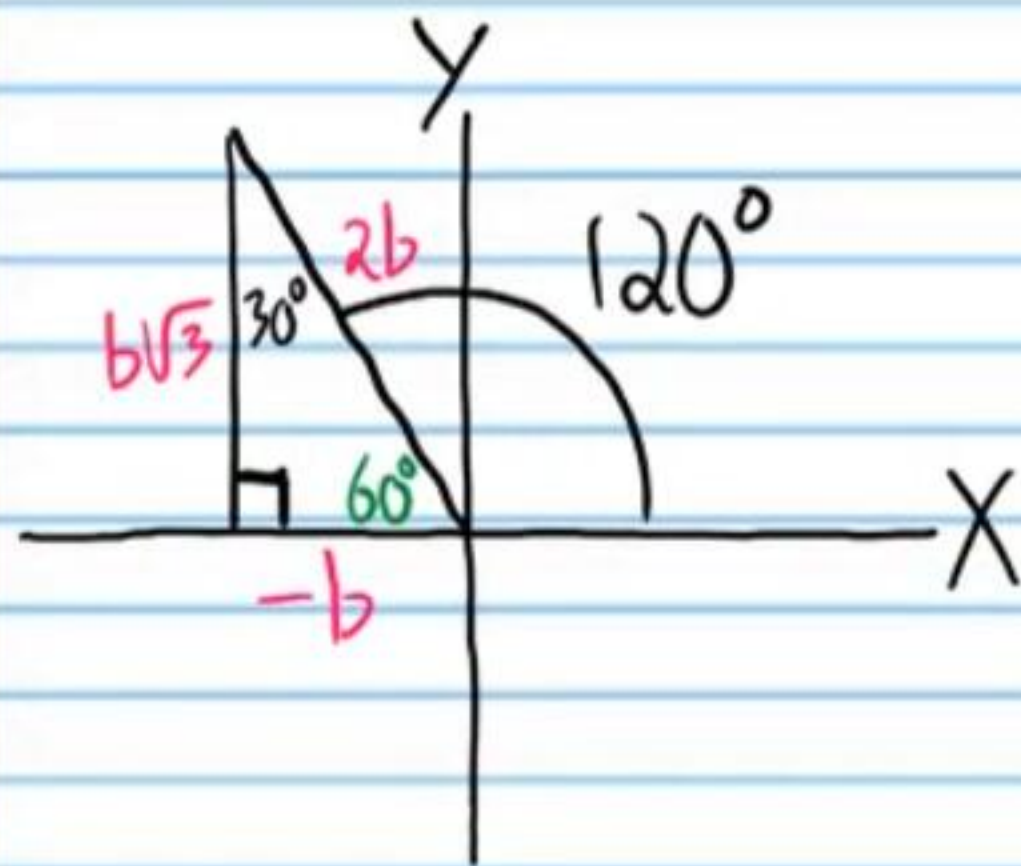


Draw each angle in standard position, write the measure of the reference angle, and draw a triangle to help find the value of the trigonometric expression. Then, find the value of the trigonometric expression *without a calculator*.

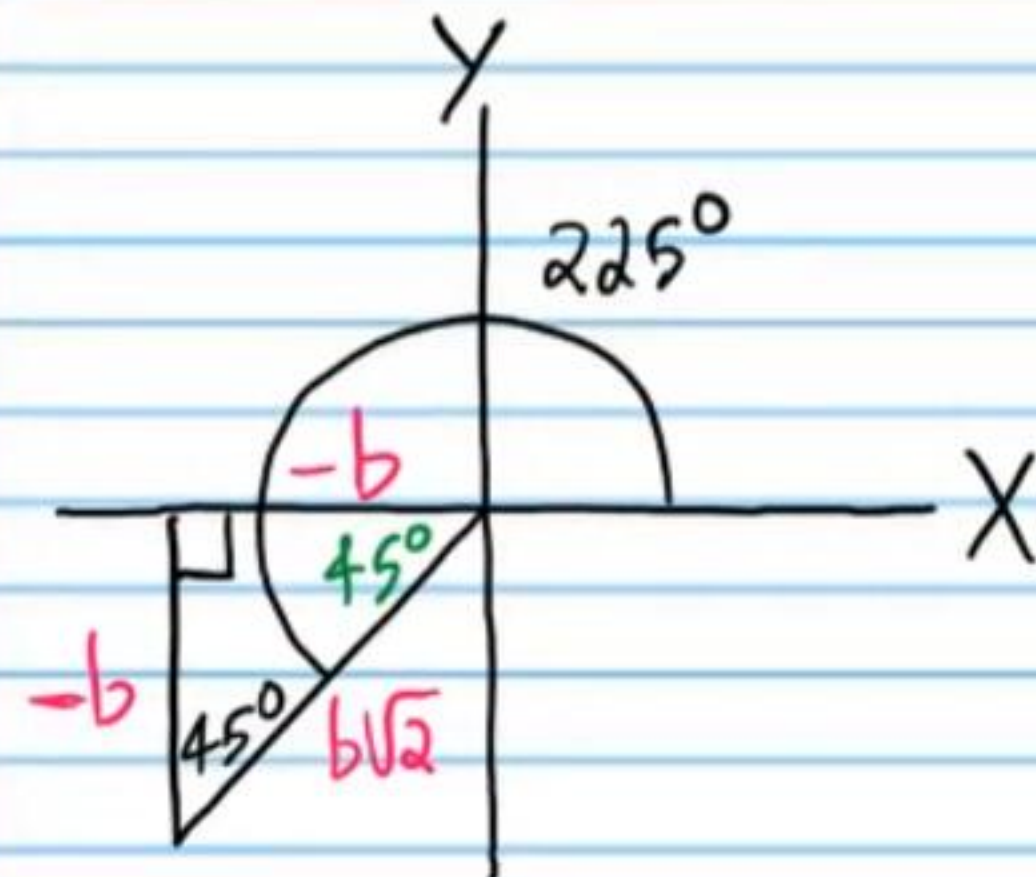
• ⑦ $\sin 45^\circ = \frac{b}{b\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2} \cdot \sqrt{2}} = \frac{\sqrt{2}}{\sqrt{4}} = \boxed{\frac{\sqrt{2}}{2}}$



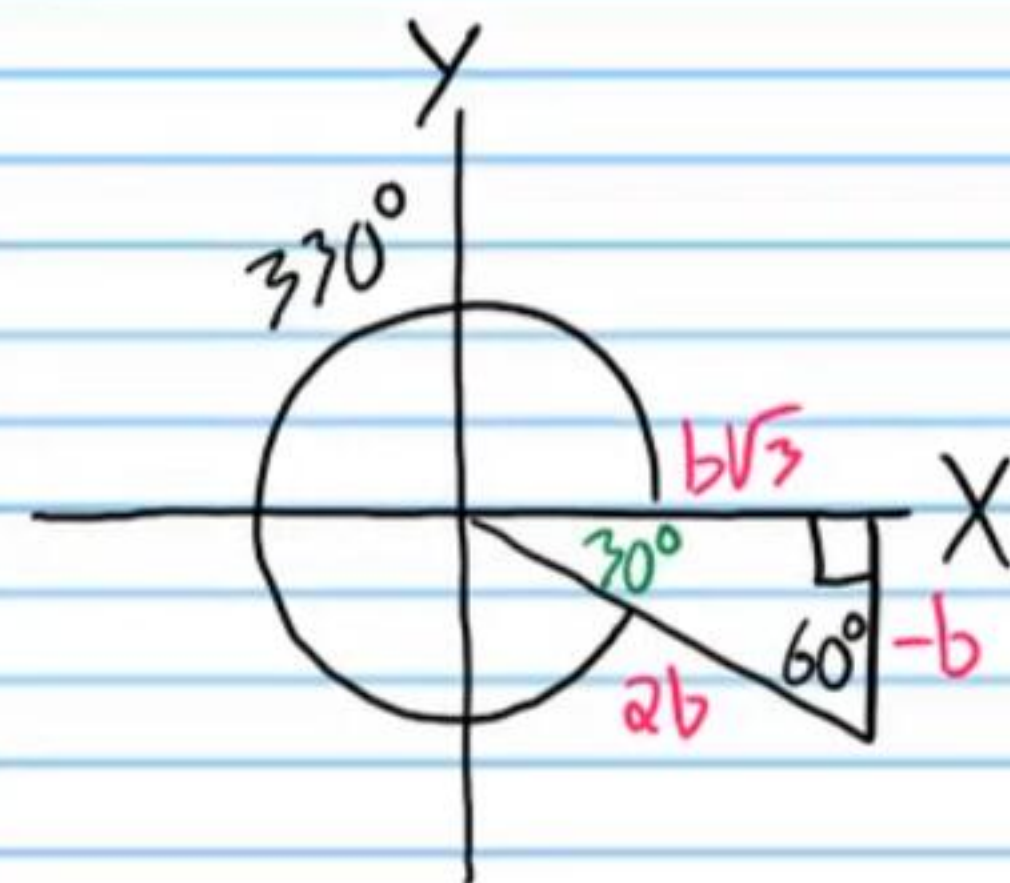
⑧ $\cos 120^\circ = \frac{-b}{2b} = \boxed{-\frac{1}{2}}$



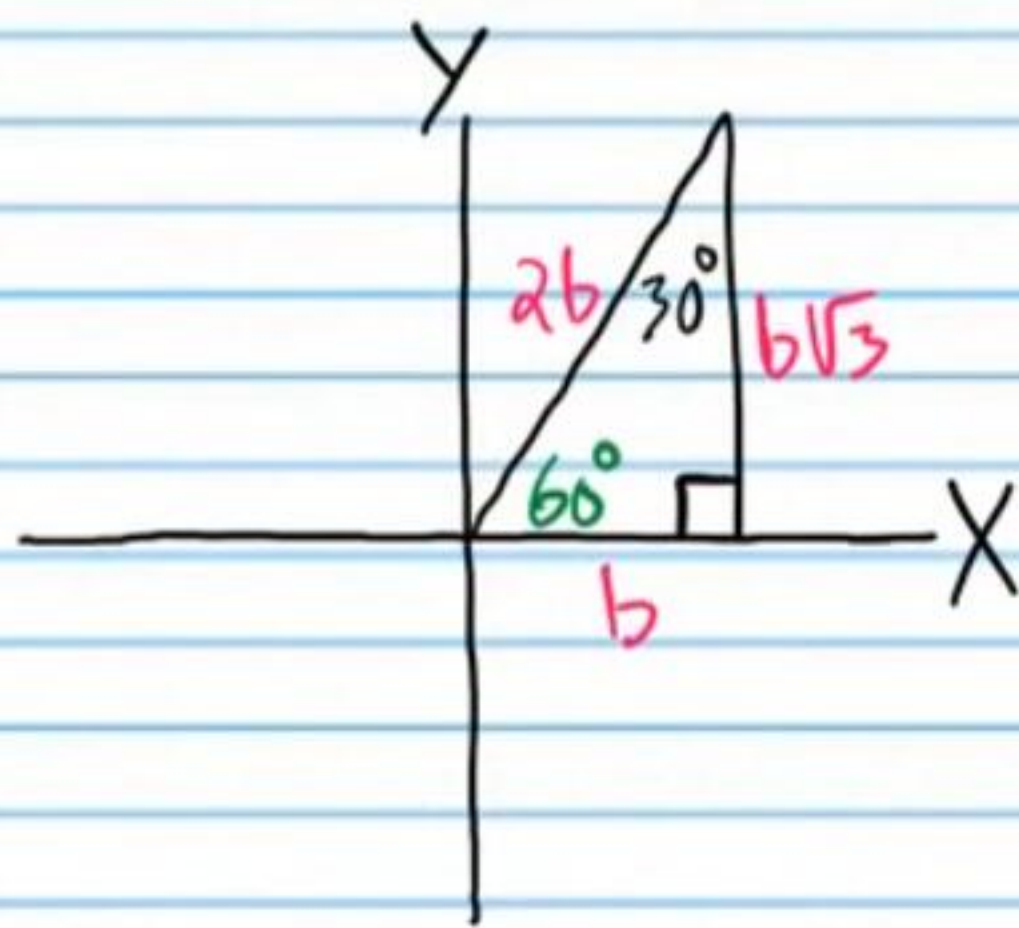
⑨ $\tan 225^\circ = \frac{-b}{-b} = \boxed{1}$



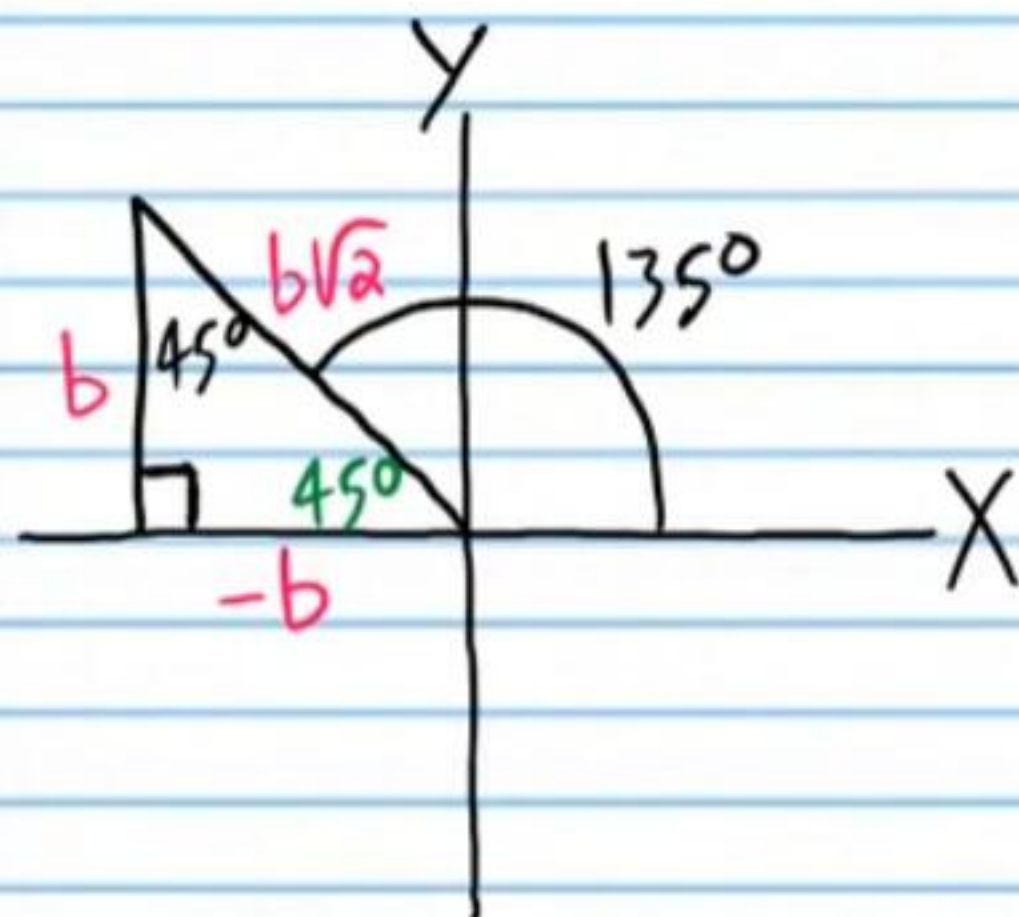
⑩ $\sec 330^\circ = \frac{2b}{b\sqrt{3}} = \frac{2}{\sqrt{3}} = \frac{2}{\sqrt{3} \cdot \sqrt{3}} = \frac{2\sqrt{3}}{\sqrt{9}} = \boxed{\frac{2\sqrt{3}}{3}}$



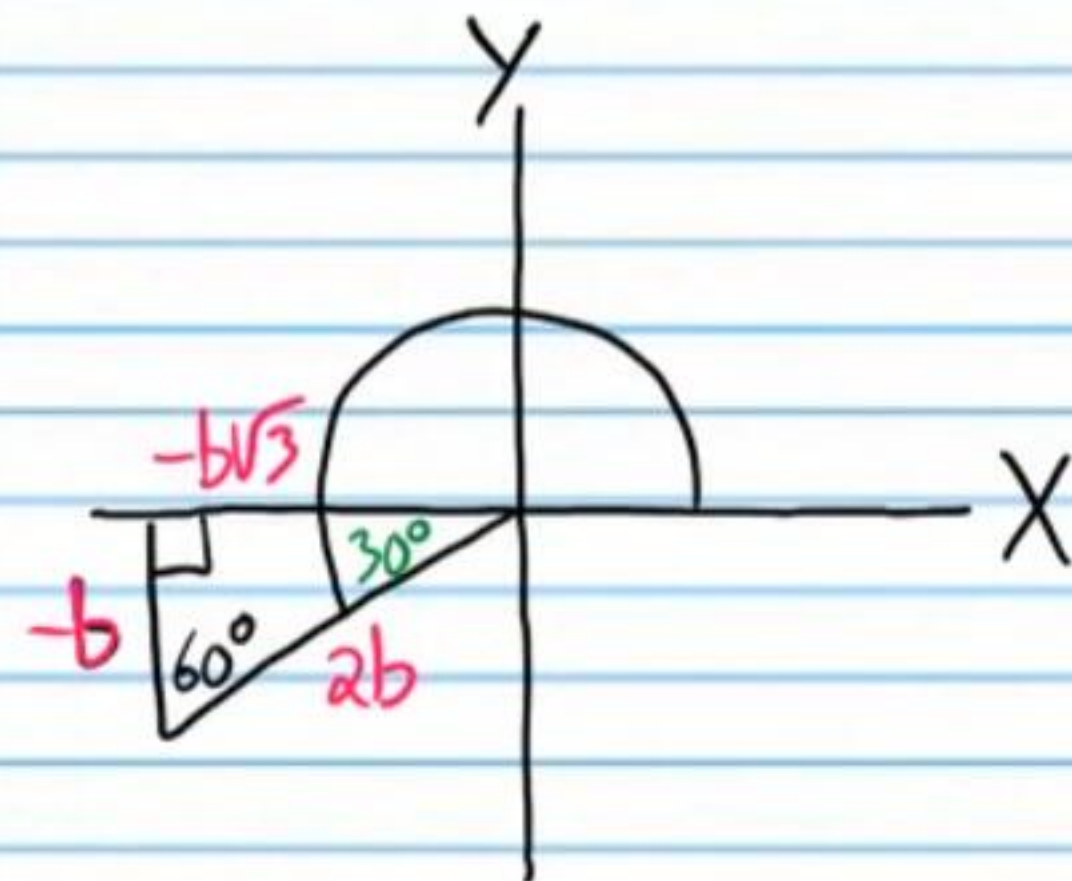
$$\bullet \textcircled{11} \cot 60^\circ = \frac{b}{b\sqrt{3}} = \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{3}}{\sqrt{9}} = \boxed{\frac{\sqrt{3}}{3}}$$



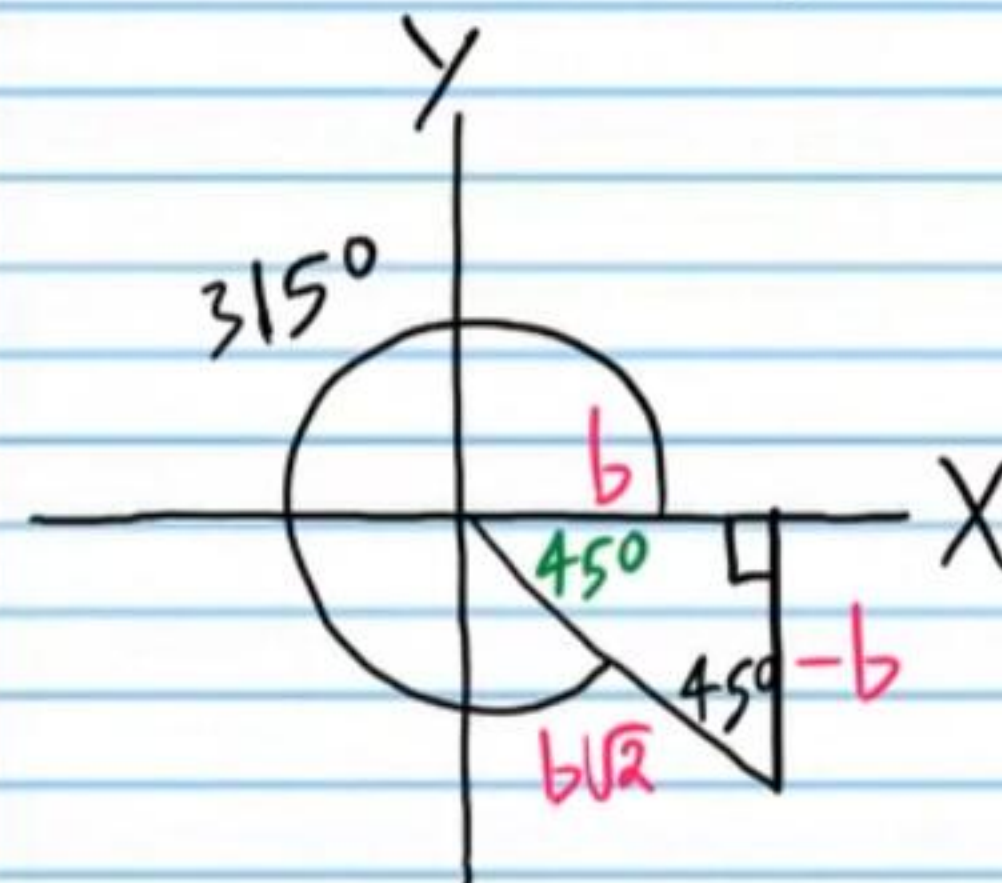
$$\textcircled{12} \sin 135^\circ = \frac{b}{b\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{4}} = \boxed{\frac{\sqrt{2}}{2}}$$



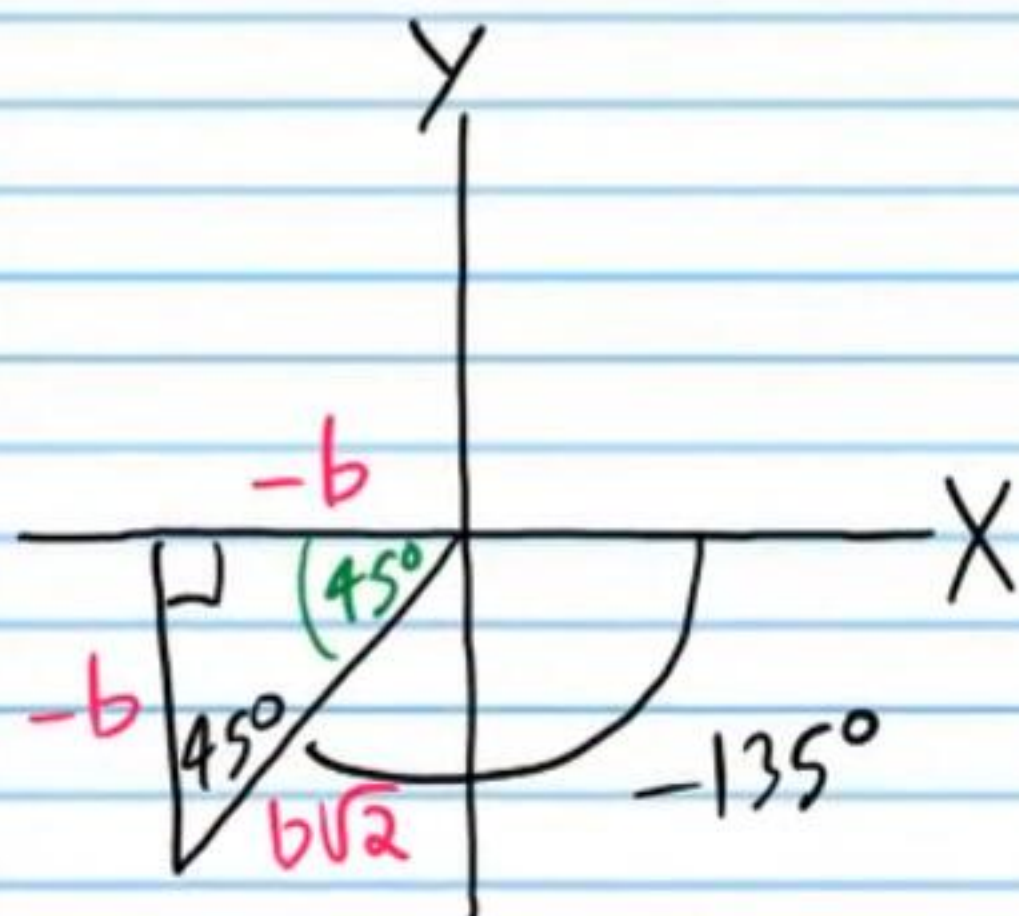
$$\textcircled{13} \csc 210^\circ = \frac{2b}{-b} = \boxed{-2}$$



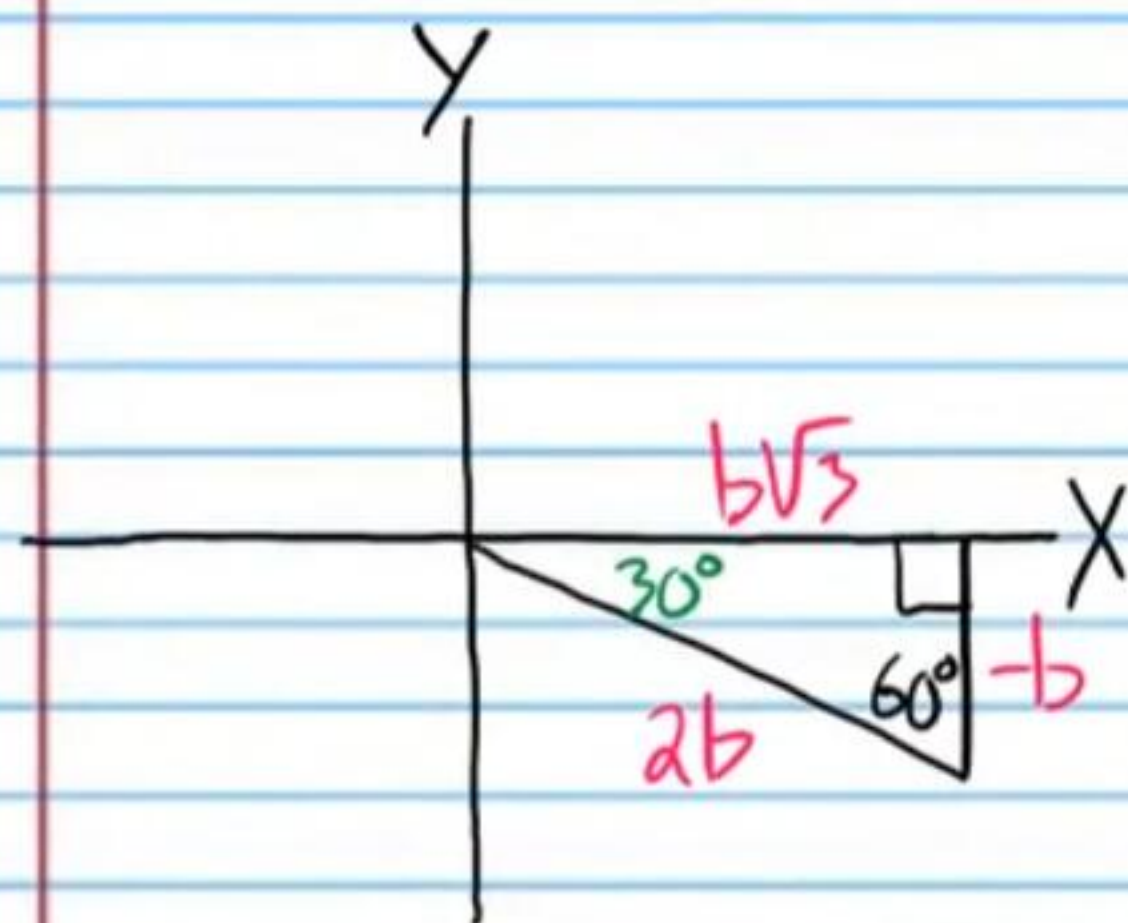
$$\textcircled{14} \tan 315^\circ = \frac{-b}{b} = \boxed{-1}$$



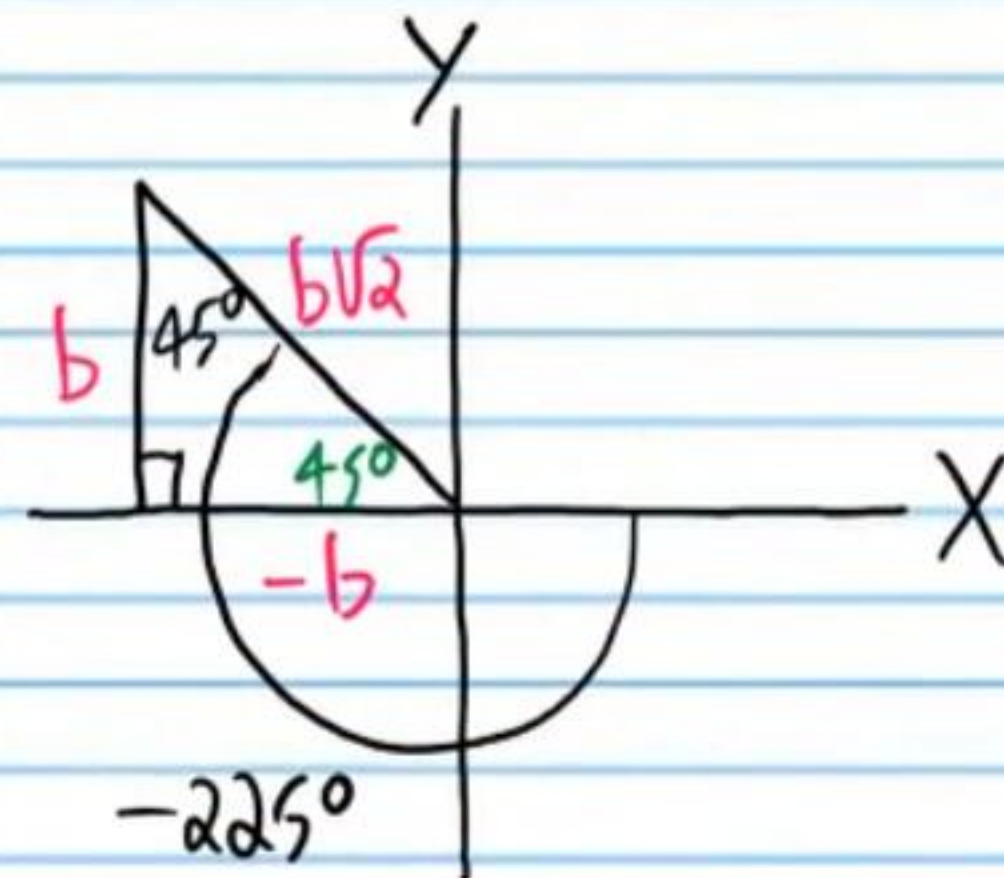
• ⑮ $\csc(-135^\circ) = \frac{b\sqrt{2}}{-b} = \boxed{-\sqrt{2}}$



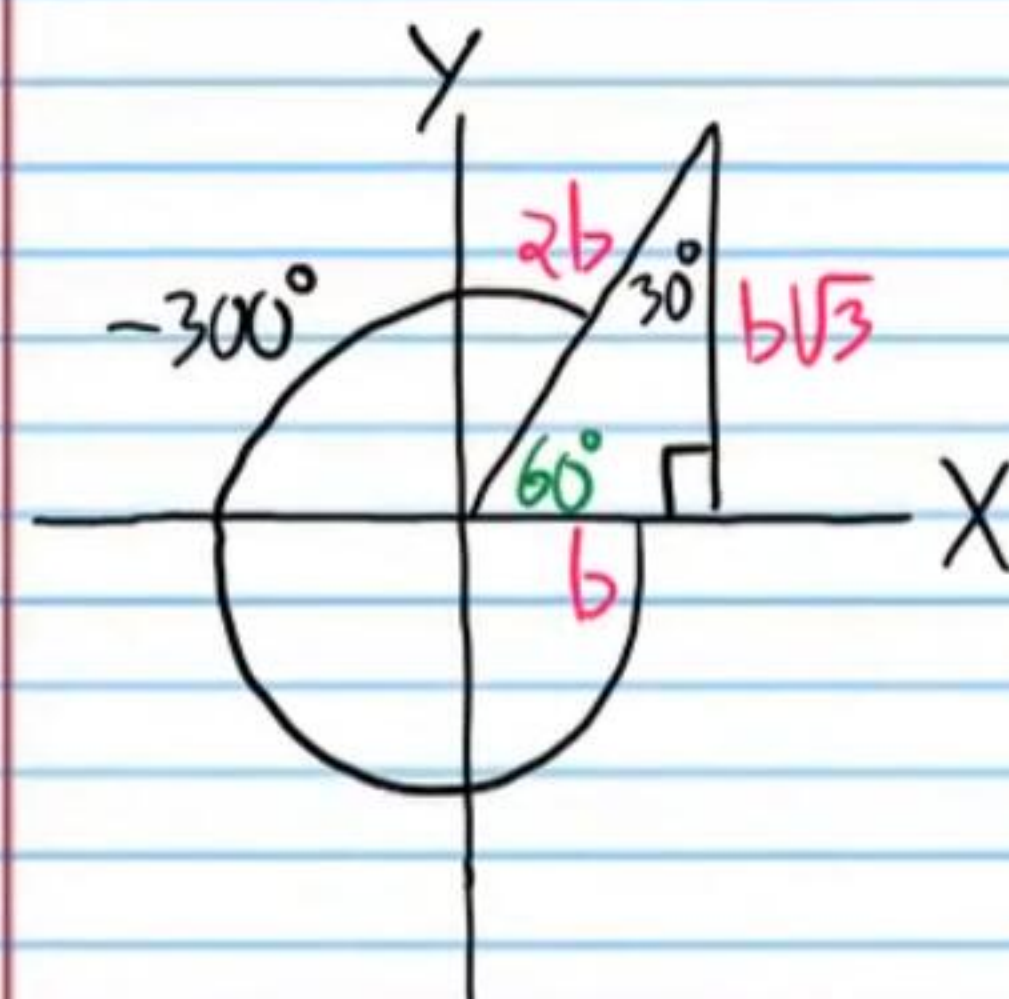
⑯ $\cot(-30^\circ) = \frac{b\sqrt{3}}{-b} = \boxed{-\sqrt{3}}$



• ⑰ $\sec(-225^\circ) = \frac{b\sqrt{2}}{-b} = \boxed{-\sqrt{2}}$

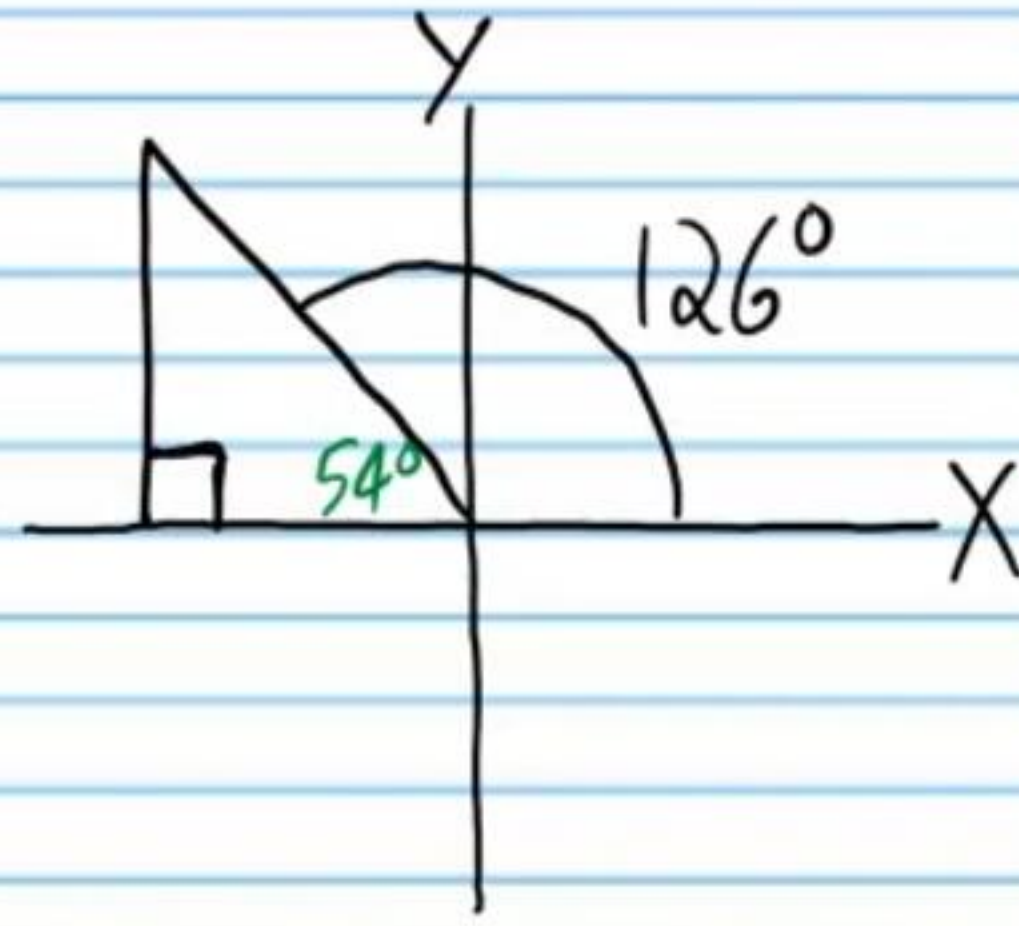
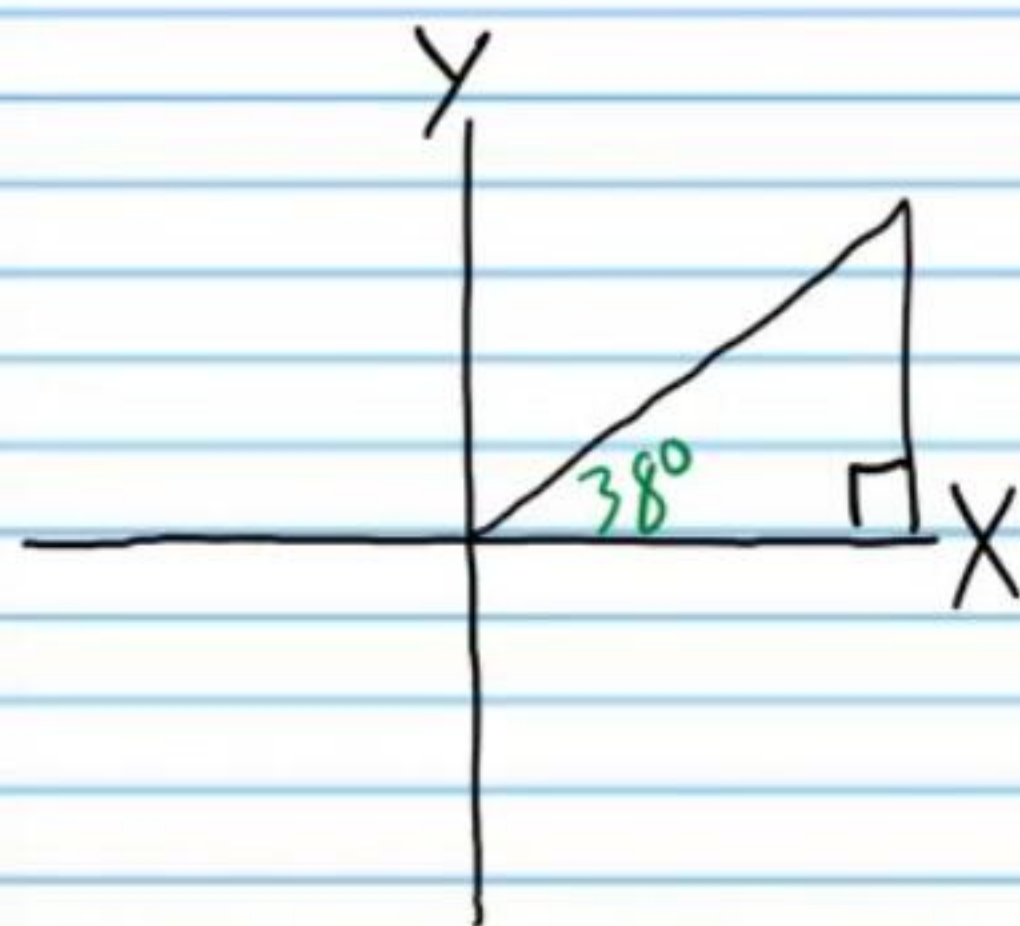


⑱ $\cos(-300^\circ) = \frac{b}{2b} = \boxed{\frac{1}{2}}$

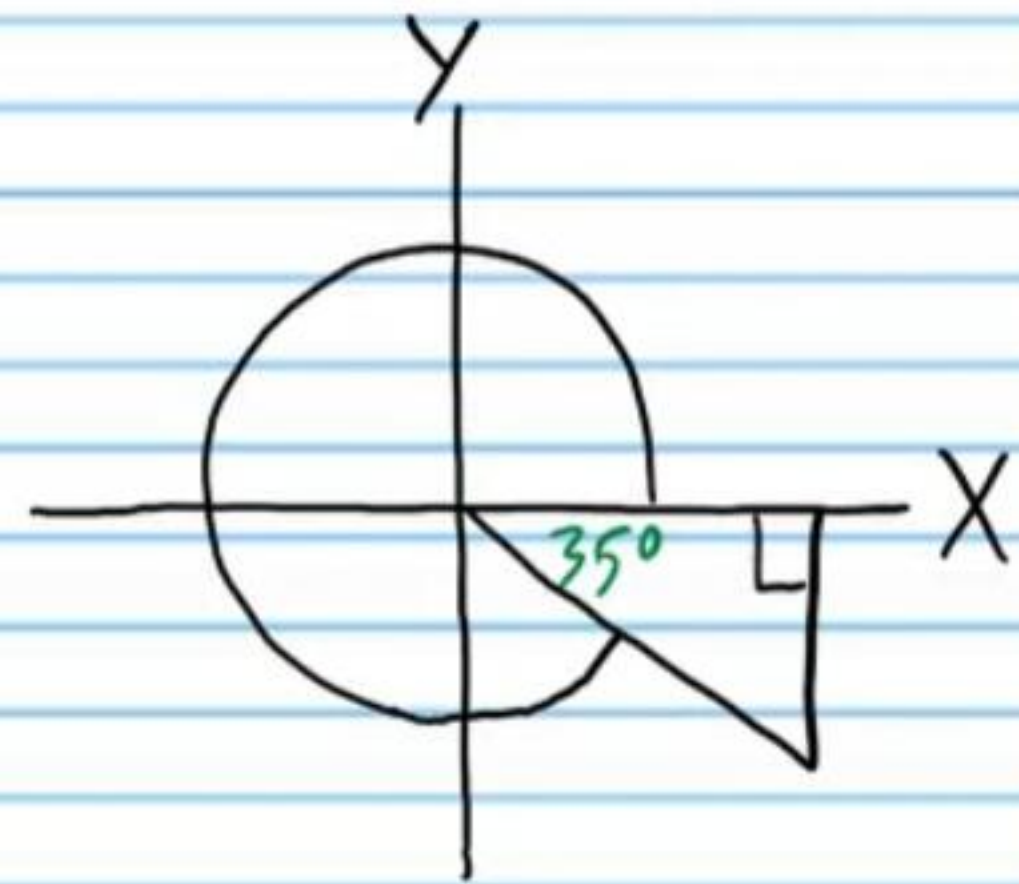
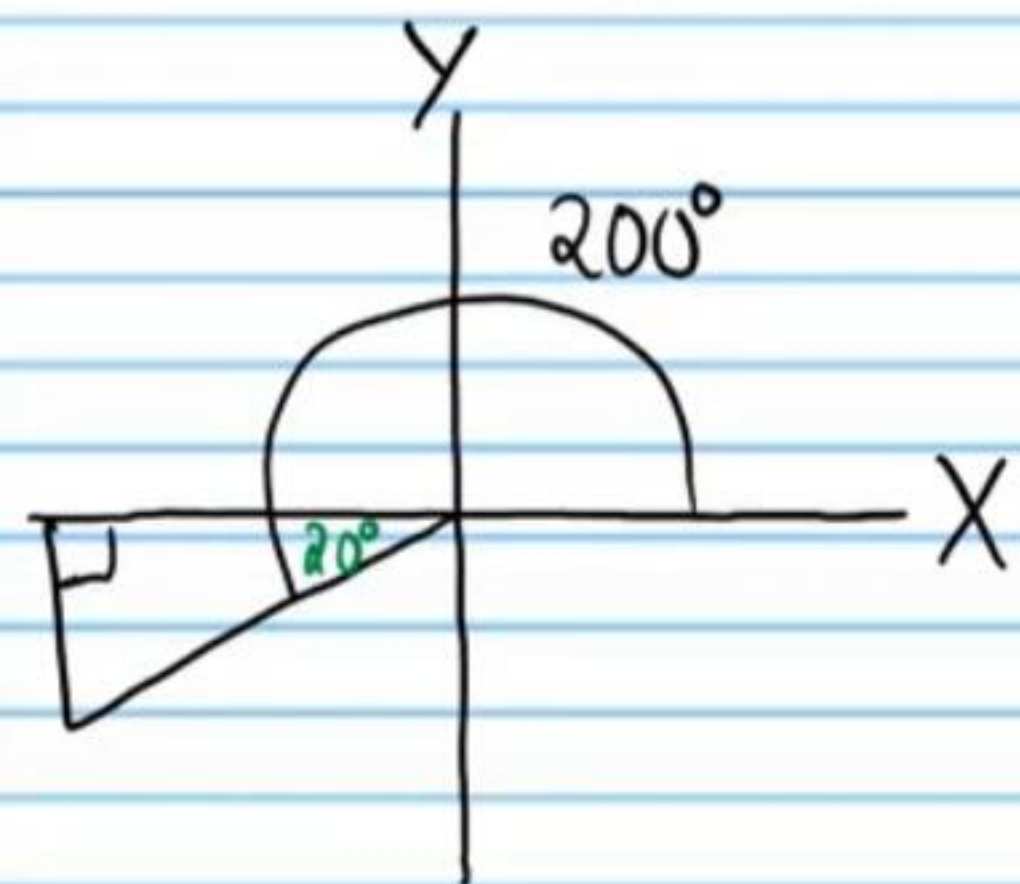


Draw each angle in standard position, write the measure of the reference angle, and draw a triangle to help understand the value of the trigonometric expression. Then, use a calculator to determine the values of each trigonometric expression. Round to four decimal places. Explain why some values are positive and some are negative.

- (19) $\cos 38^\circ \approx \boxed{0.788}$ (20) $\tan 126^\circ \approx \boxed{-1.3764}$

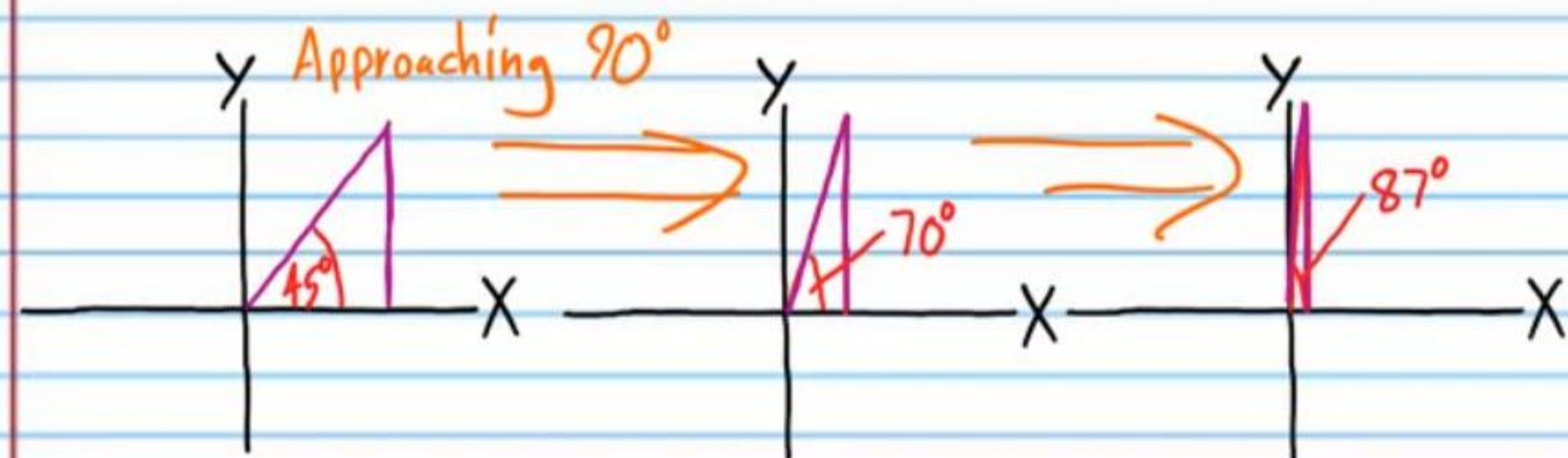


- (21) $\sin 200^\circ \approx \boxed{-0.342}$ (22) $\cos 325^\circ \approx \boxed{0.8192}$

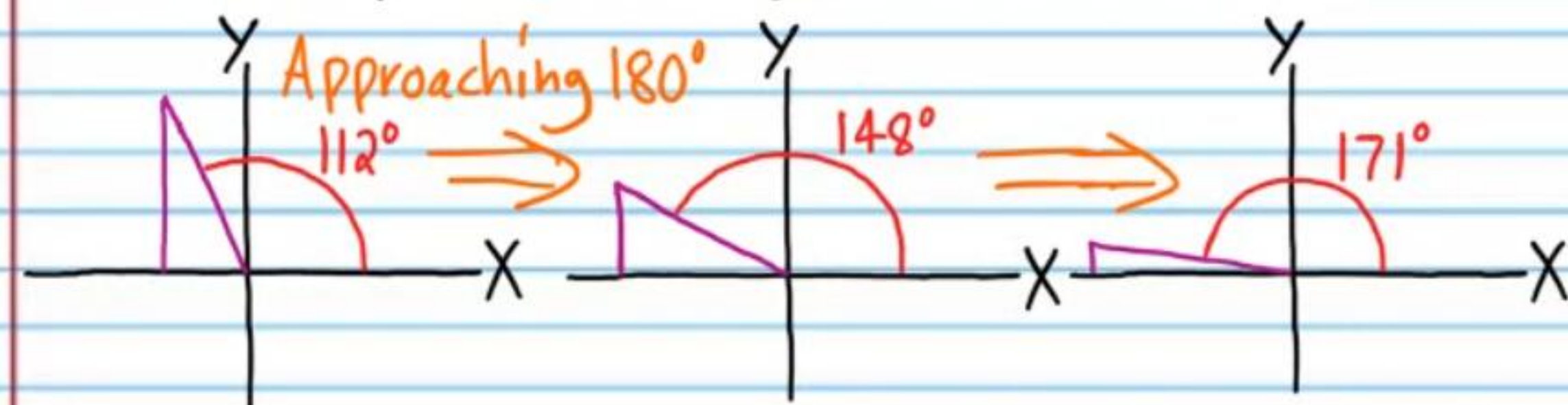


Collapsed Triangles ($\theta = 90K$, K any integer)

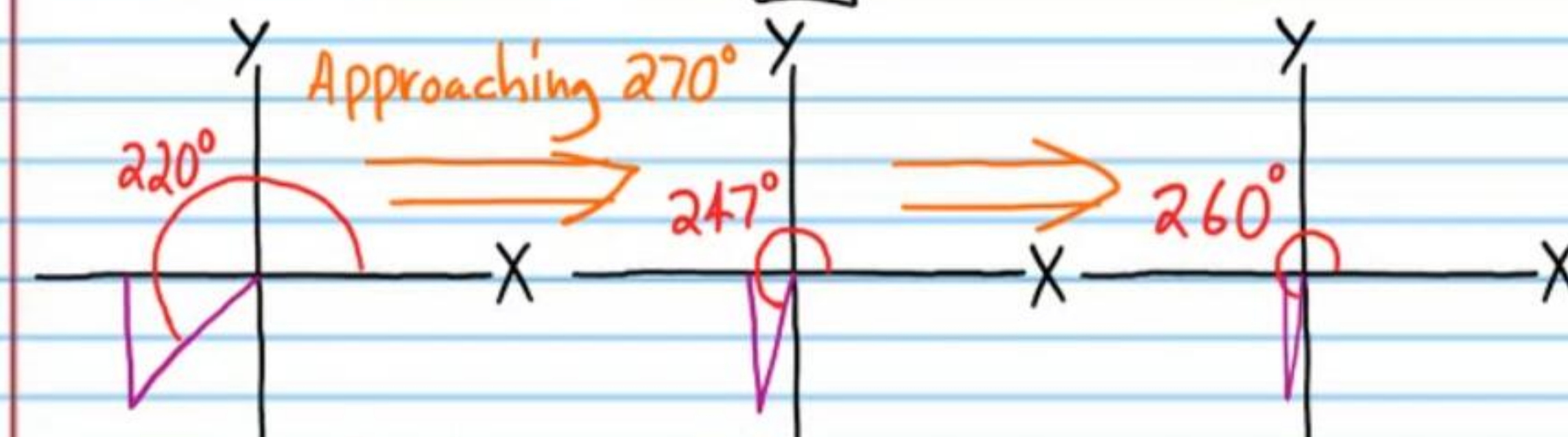
Example I) $\sin 90^\circ = \boxed{1}$



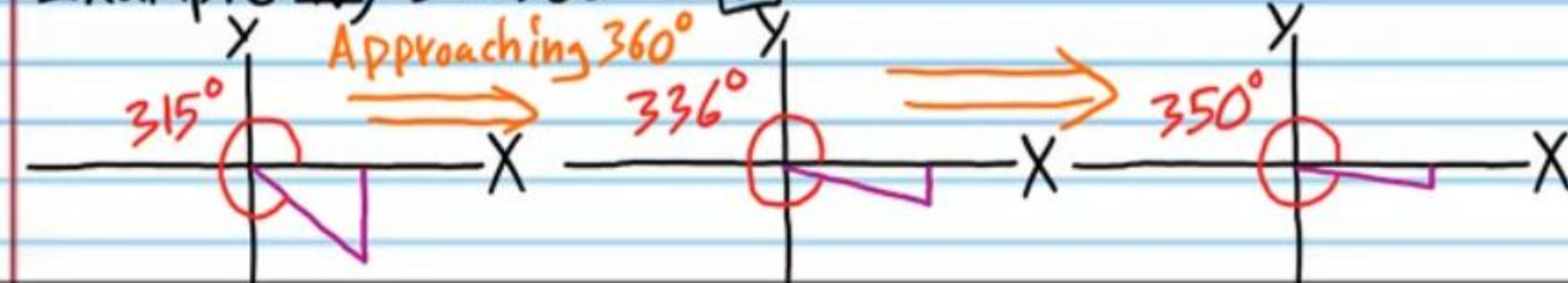
Example II) $\sin 180^\circ = \boxed{0}$



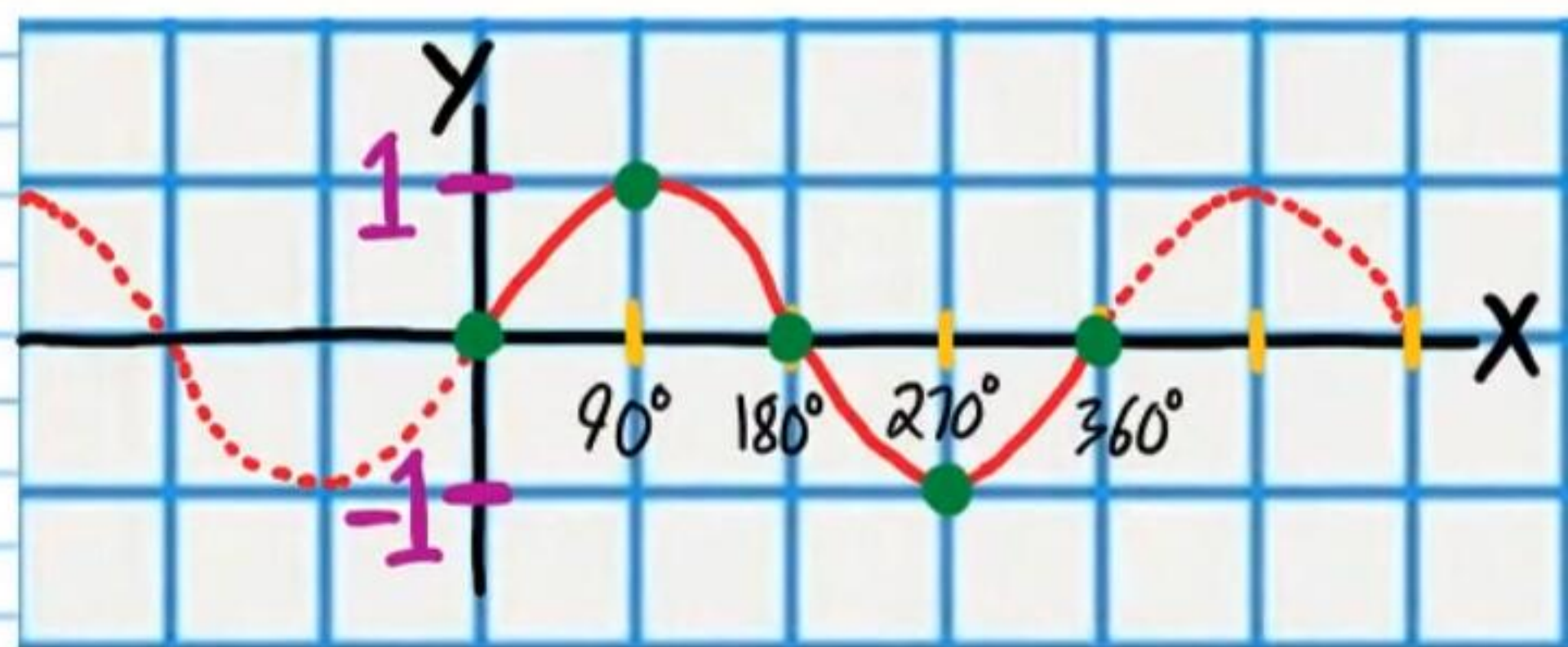
Example III) $\sin 270^\circ = \boxed{-1}$



Example IV) $\sin 360^\circ = \boxed{0}$



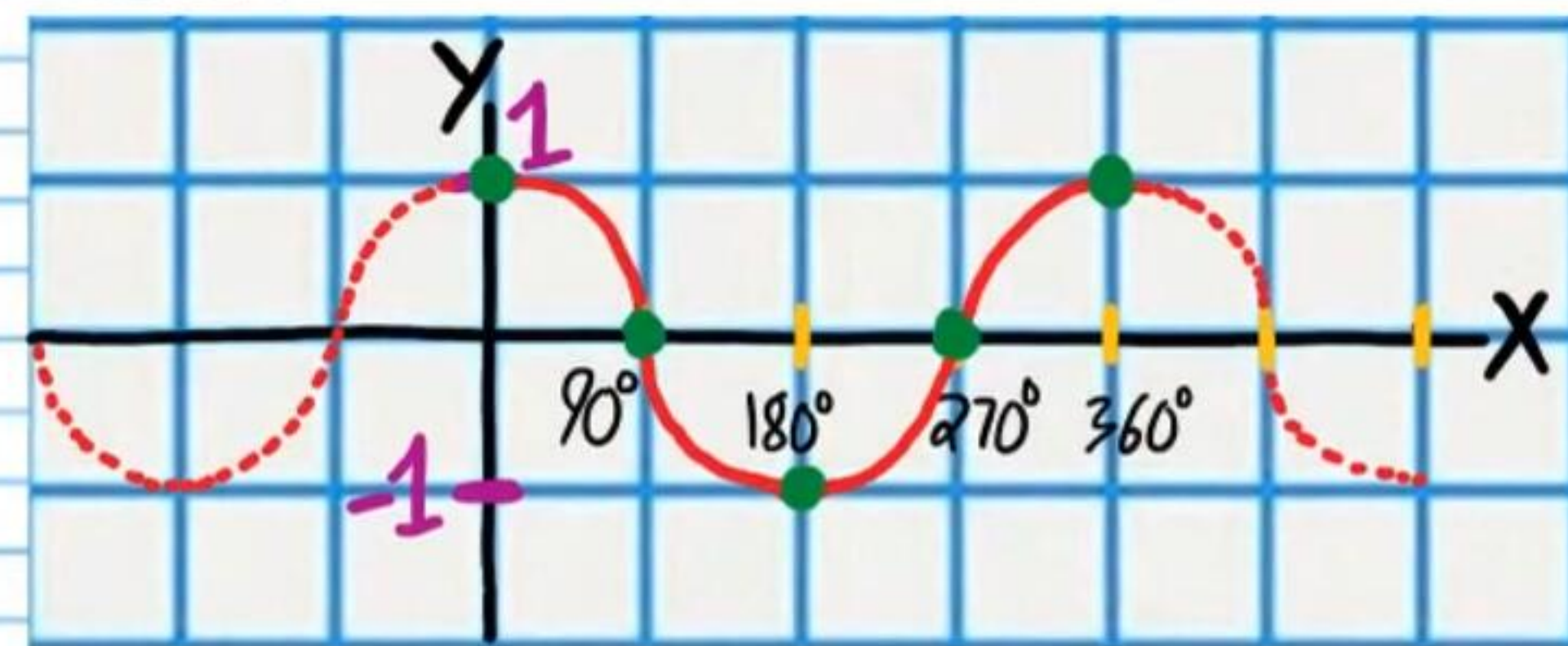
$$y = \sin x$$



Use the graph above to find the following values.

- (23) $\sin 270^\circ = \boxed{-1}$ (24) $\sin 360^\circ = \boxed{0}$ (25) $\sin 0^\circ = \boxed{0}$
 (26) $\sin(-90^\circ) = \boxed{-1}$ (27) $\sin(-180^\circ) = \boxed{0}$ (28) $\sin 450^\circ = \boxed{1}$

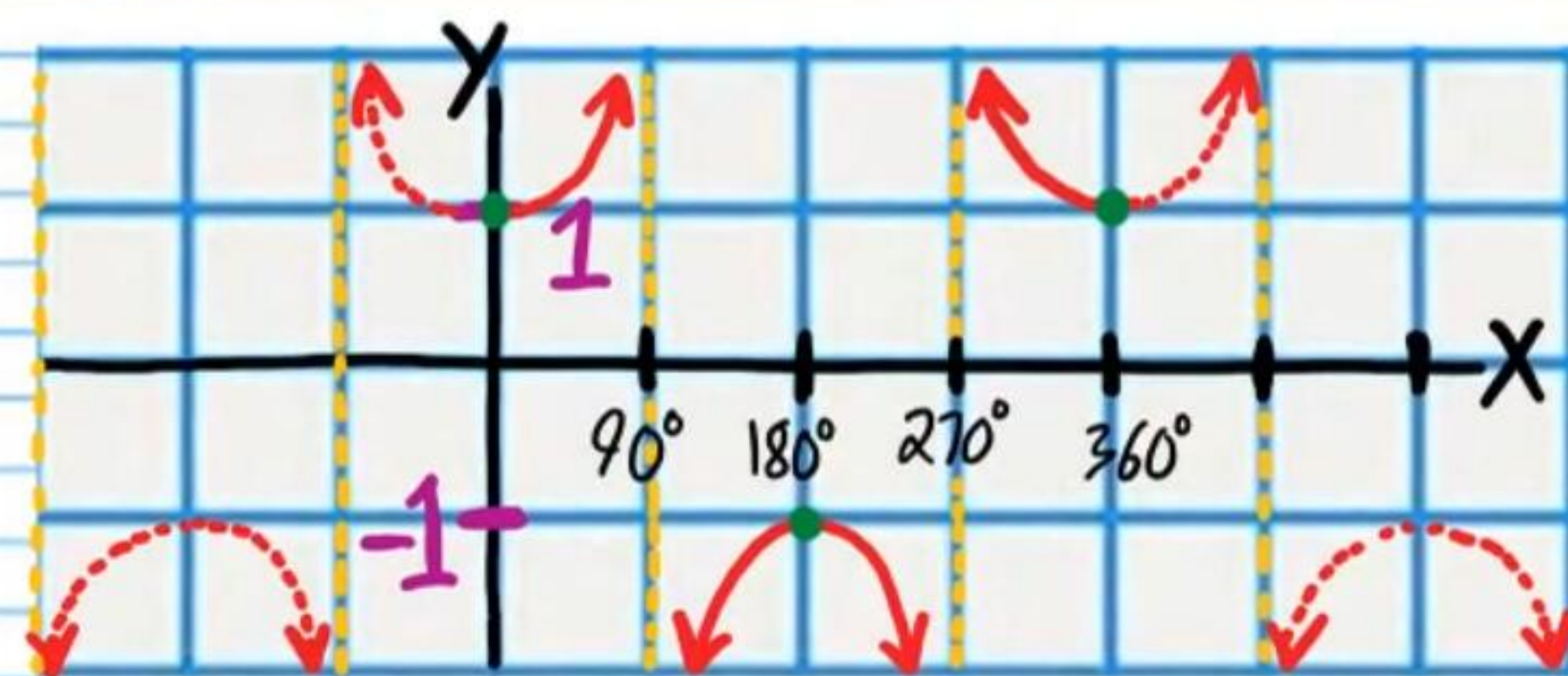
$$y = \cos x$$



Use the graph above to find the following values.

- (29) $\cos 90^\circ = \boxed{0}$ (30) $\cos 180^\circ = \boxed{-1}$ (31) $\cos 270^\circ = \boxed{0}$
 (32) $\cos 360^\circ = \boxed{1}$ (33) $\cos 0^\circ = \boxed{1}$ (34) $\cos(-90^\circ) = \boxed{0}$

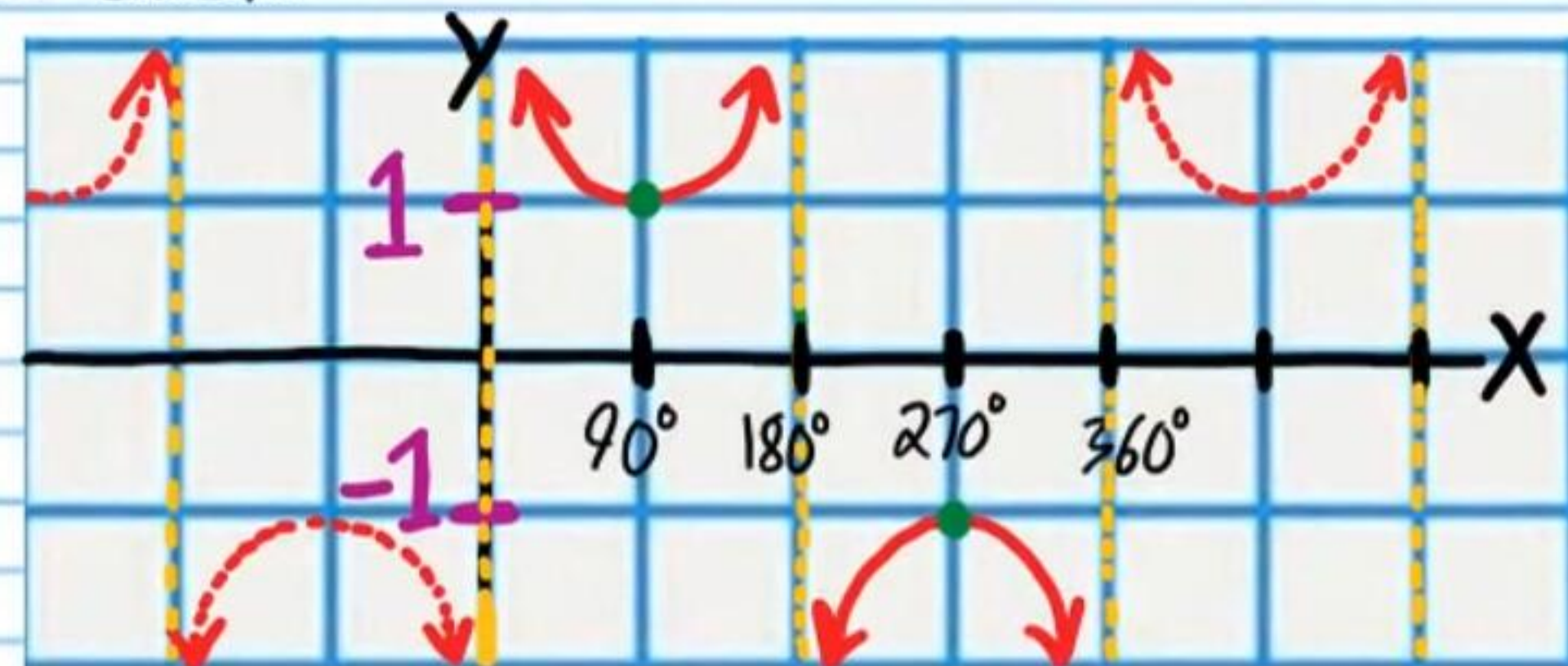
$$y = \sec x$$



Use the graph above to find the following values.

- (35) $\sec(180^\circ) = \boxed{-1}$ (36) $\sec 90^\circ = \boxed{\text{und}}$ (37) $\sec 270^\circ = \boxed{\text{und}}$
 (38) $\sec(-90^\circ) = \boxed{\text{und}}$ (39) $\sec 0^\circ = \boxed{1}$ (40) $\sec(-180^\circ) = \boxed{-1}$

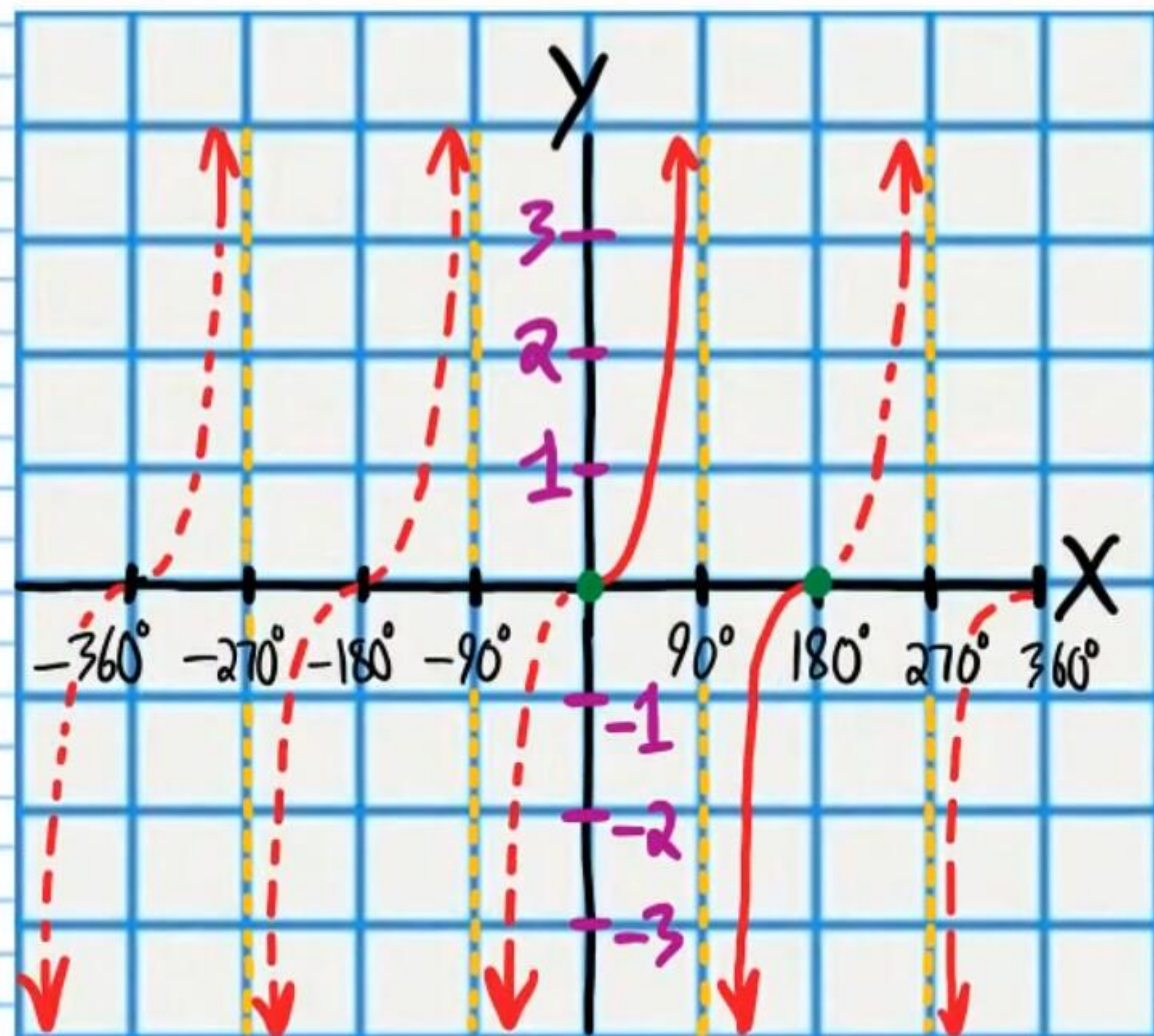
$$y = \csc x$$



Use the graph above to find the following values.

- (41) $\csc 90^\circ = \boxed{1}$ (42) $\csc 180^\circ = \boxed{\text{und}}$ (43) $\csc(270^\circ) = \boxed{-1}$
 (44) $\csc 0^\circ = \boxed{\text{und}}$ (45) $\csc(-90^\circ) = \boxed{-1}$ (46) $\csc 360^\circ = \boxed{\text{und}}$

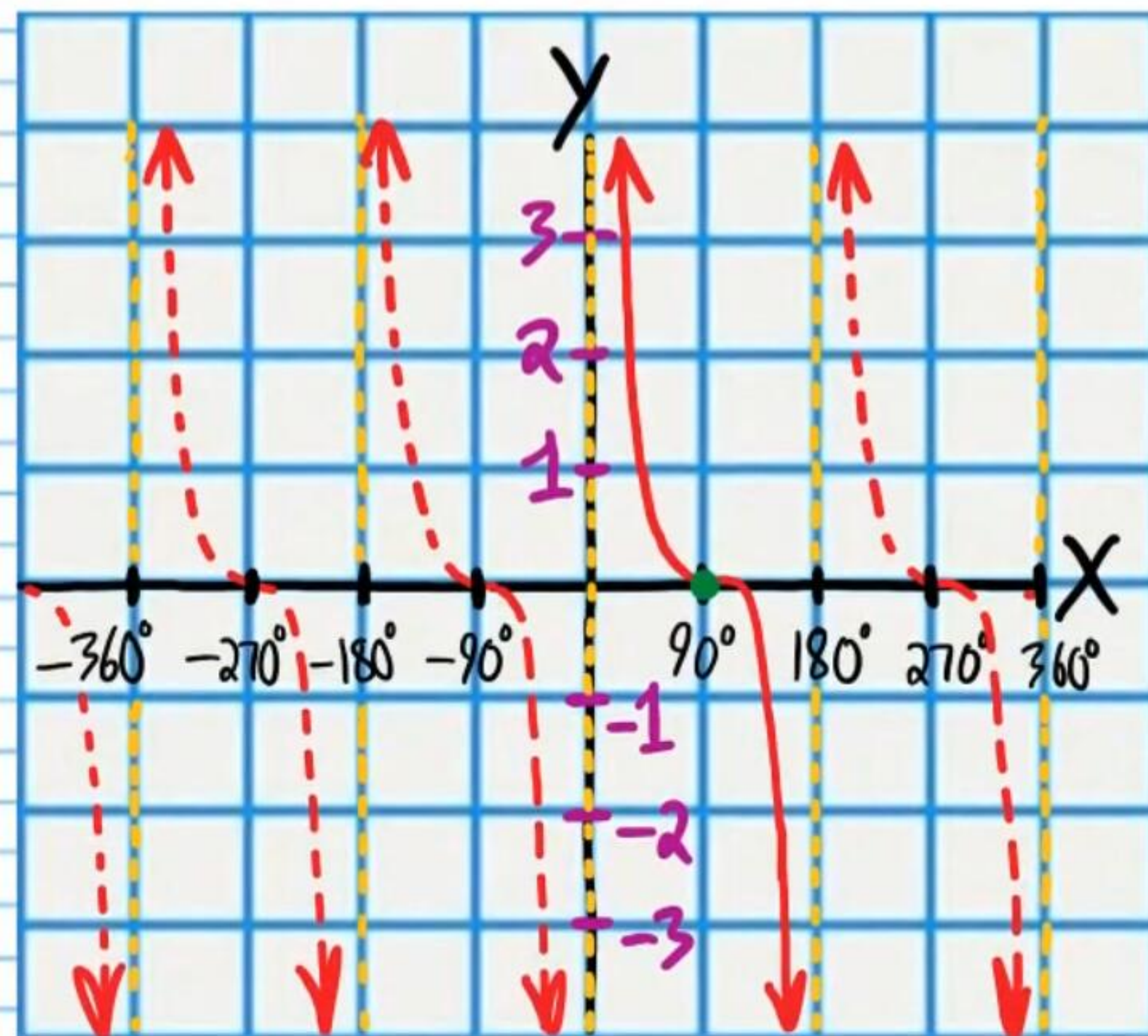
$$y = \tan x$$



Use the graph above to find the following values.

- (47) $\tan 180^\circ = \boxed{0}$ (48) $\tan 90^\circ = \boxed{\text{und}}$ (49) $\tan 0^\circ = \boxed{0}$
- (50) $\tan(-270^\circ) = \boxed{\text{und}}$ (51) $\tan(-180^\circ) = \boxed{0}$
- (52) $\tan(360^\circ) = \boxed{0}$

$$y = \cot x$$



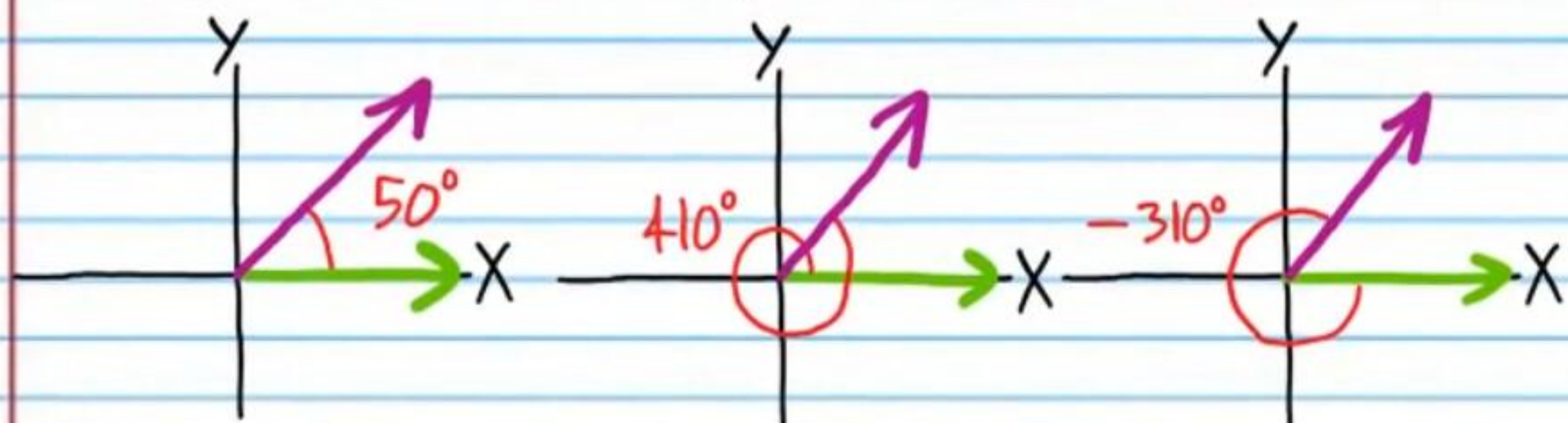
Use the graph above to find the following values.

- (53) $\cot 90^\circ = \boxed{0}$ (54) $\cot 180^\circ = \boxed{\text{und}}$
- (55) $\cot(-90^\circ) = \boxed{0}$ (56) $\cot 0^\circ = \boxed{\text{und}}$
- (57) $\cot 270^\circ = \boxed{0}$

Coterminal Angles

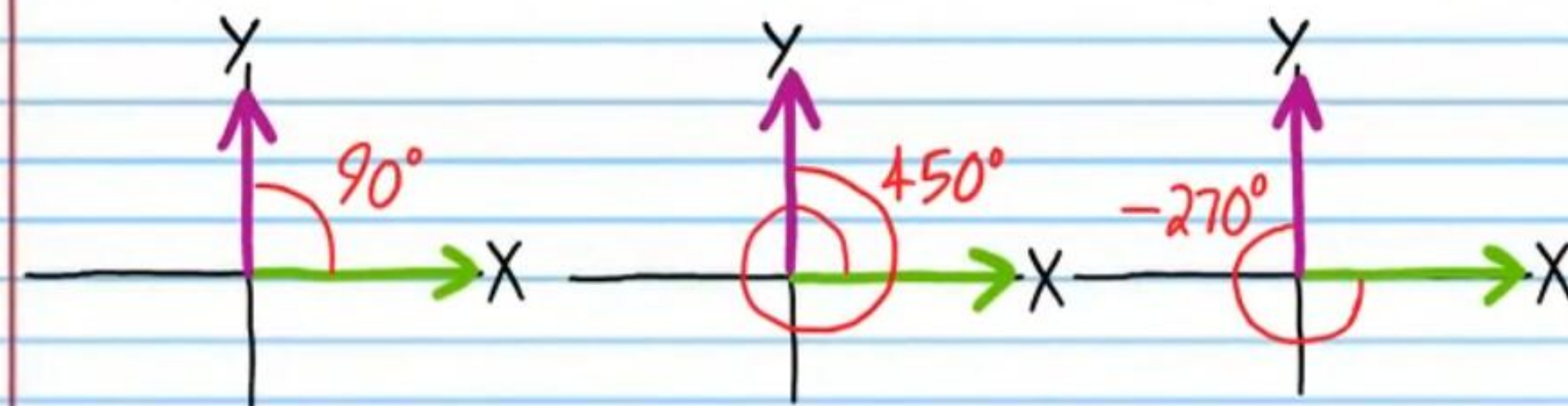
Coterminal Angles: Two or more angles in standard position whose terminal sides have the same location.

Example I

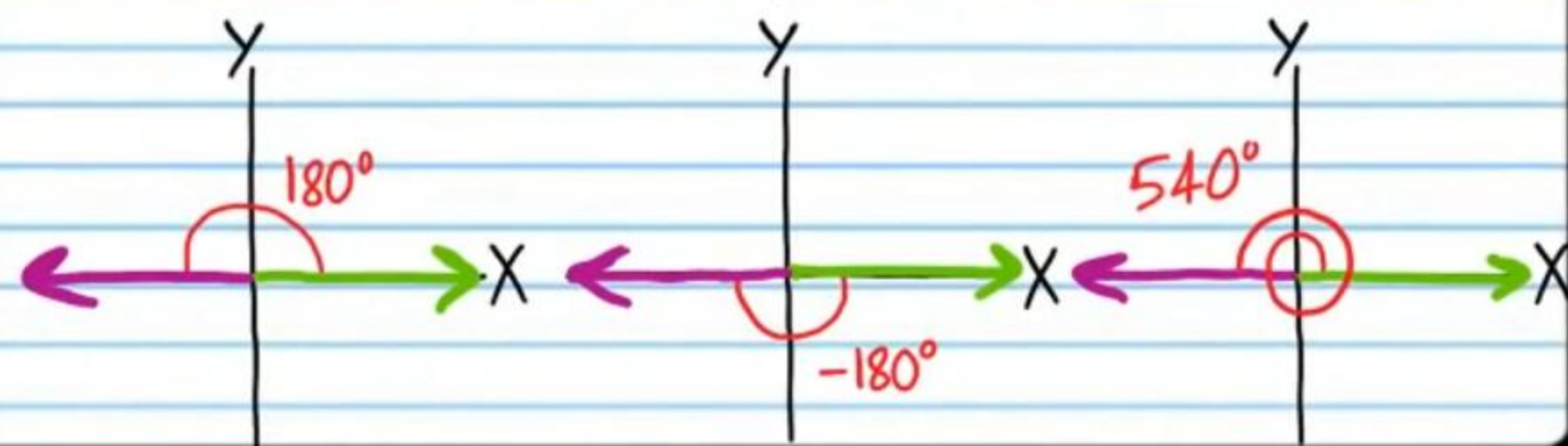


Given an angle in standard position, the measures of coterminal angles can be found by adding or subtracting multiples of 360° to the measure of the original angle.

Example II



Example III



Practice

The measures of angles in standard position are written below. Find the measures of three coterminal angles in standard position.

• (58) 200°

$$\theta_1 = 200^\circ + 360^\circ = \boxed{560^\circ}$$

$$\theta_2 = 200^\circ + 360^\circ + 360^\circ = \boxed{920^\circ}$$

$$\theta_3 = 200^\circ - 360^\circ = \boxed{-160^\circ}$$

(59) -1°

$$\theta_1 = -1^\circ + 360^\circ = \boxed{359^\circ}$$

$$\theta_2 = -1^\circ + 360^\circ + 360^\circ = \boxed{719^\circ}$$

$$\theta_3 = -1^\circ - 360^\circ = \boxed{-361^\circ}$$

• (60) 80°

$$\theta_1 = 80^\circ + 360^\circ = \boxed{440^\circ}$$

$$\theta_2 = 80^\circ + 360^\circ + 360^\circ = \boxed{800^\circ}$$

$$\theta_3 = 80^\circ - 360^\circ = \boxed{-280^\circ}$$

$$\theta_4 = 80^\circ - 360^\circ - 360^\circ = \boxed{-800^\circ}$$

(61) 0°

$$\theta_1 = 0^\circ + 360^\circ = \boxed{360^\circ}$$

$$\theta_2 = 0^\circ + 360^\circ + 360^\circ = \boxed{720^\circ}$$

$$\theta_3 = 0^\circ - 360^\circ = \boxed{-360^\circ}$$

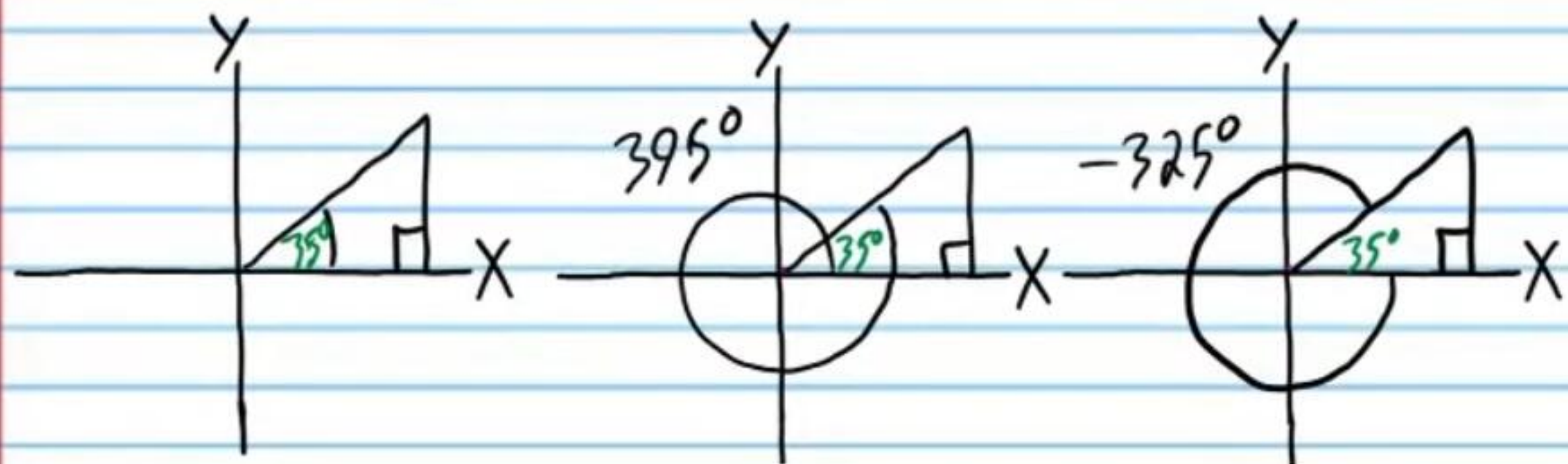
$$\theta_4 = 0^\circ - 360^\circ - 360^\circ = \boxed{-720^\circ}$$

Trigonometric Values of Coterminal Angles

Use a calculator to show that the following statements are true. Then, draw a reference triangle in standard position to show why each statement is true.

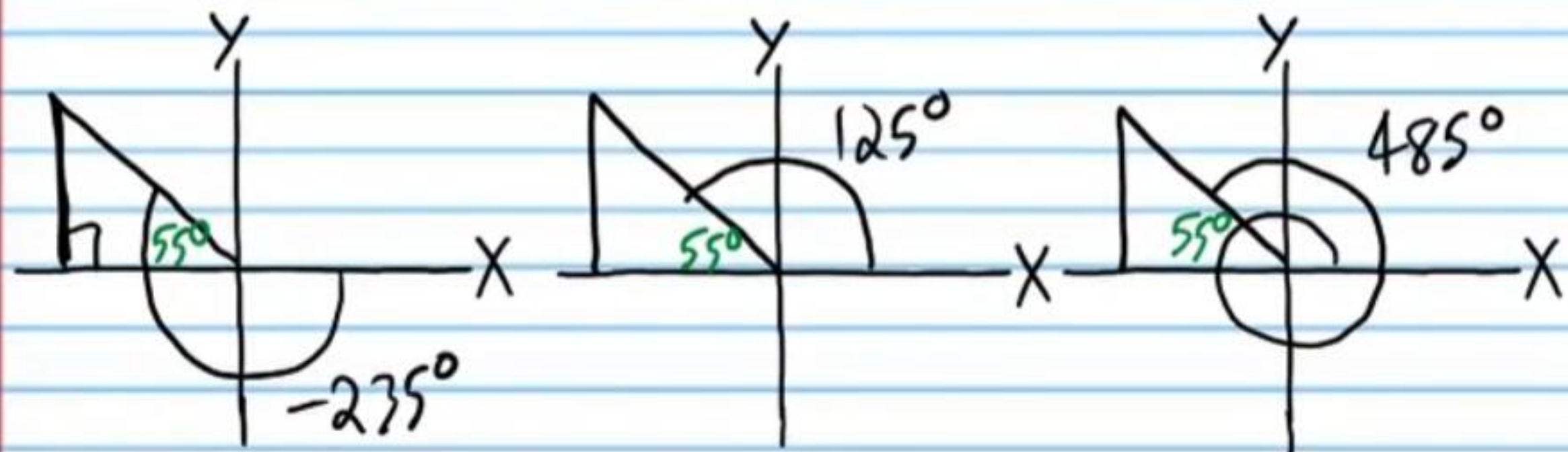
• (62) $\sin 35^\circ = \sin 395^\circ = \sin(-325^\circ)$

$$0.5736 = 0.5736 = 0.5736$$



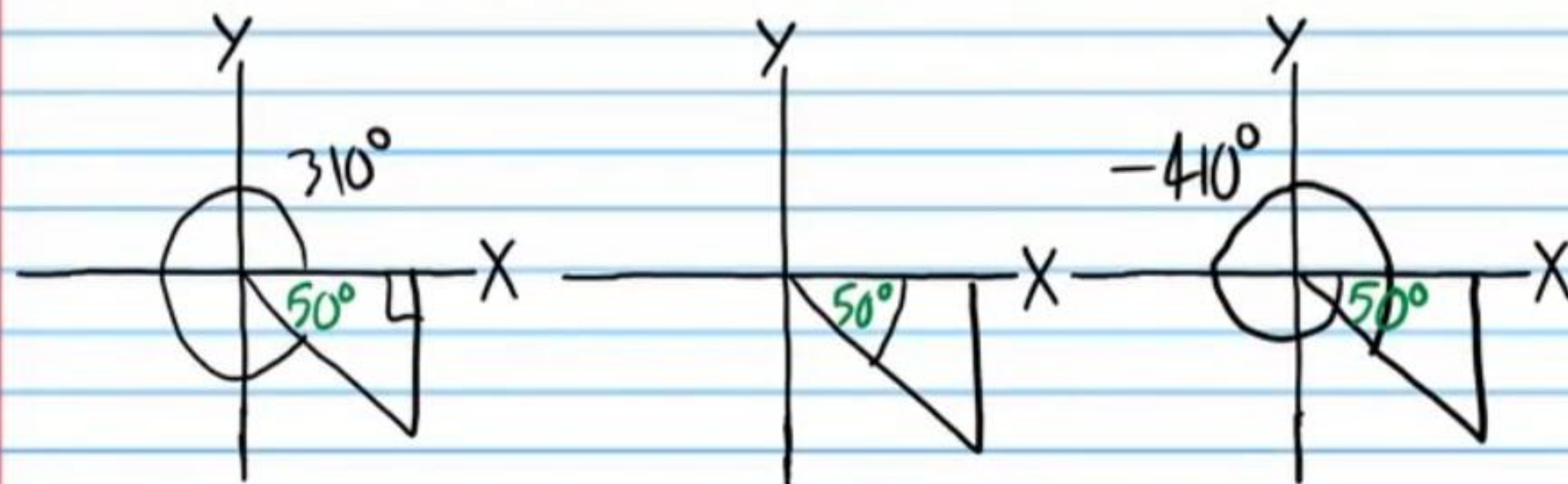
(63) $\tan(-235^\circ) = \tan 125^\circ = \tan 485^\circ$

$$-1.4281 = -1.4281 = -1.4281$$



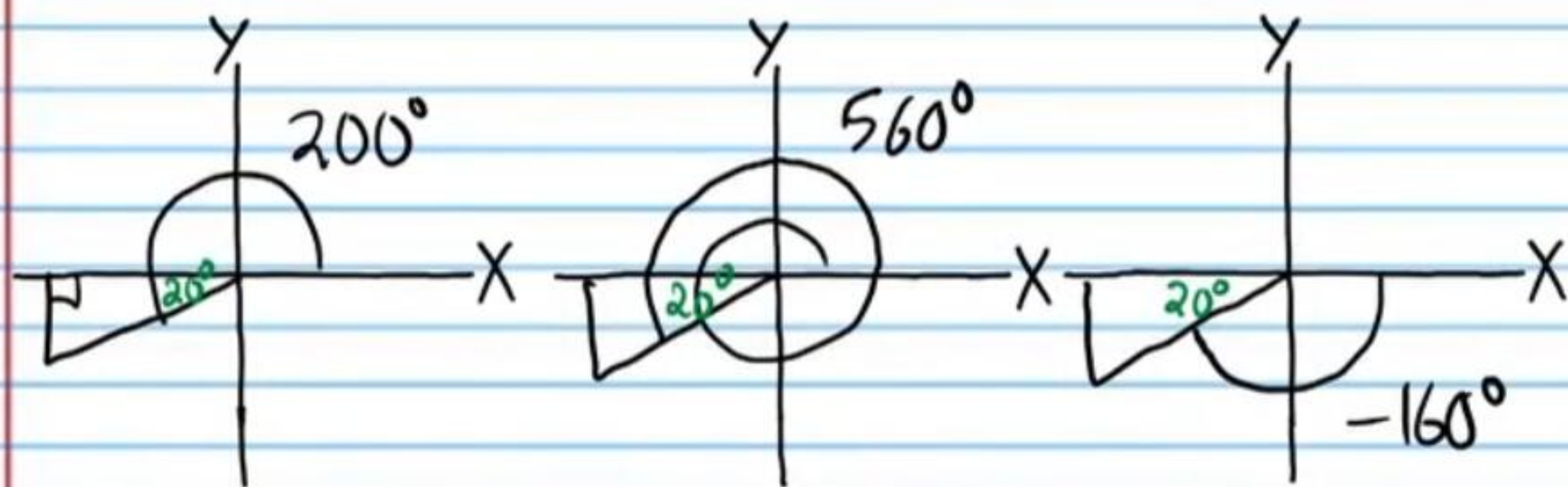
• (64) $\cos 310^\circ = \cos(-50^\circ) = \cos(-410^\circ)$

$$0.6428 = 0.6428 = 0.6428$$



(65) $\sin 200^\circ = \sin 560^\circ = \sin(-160^\circ)$

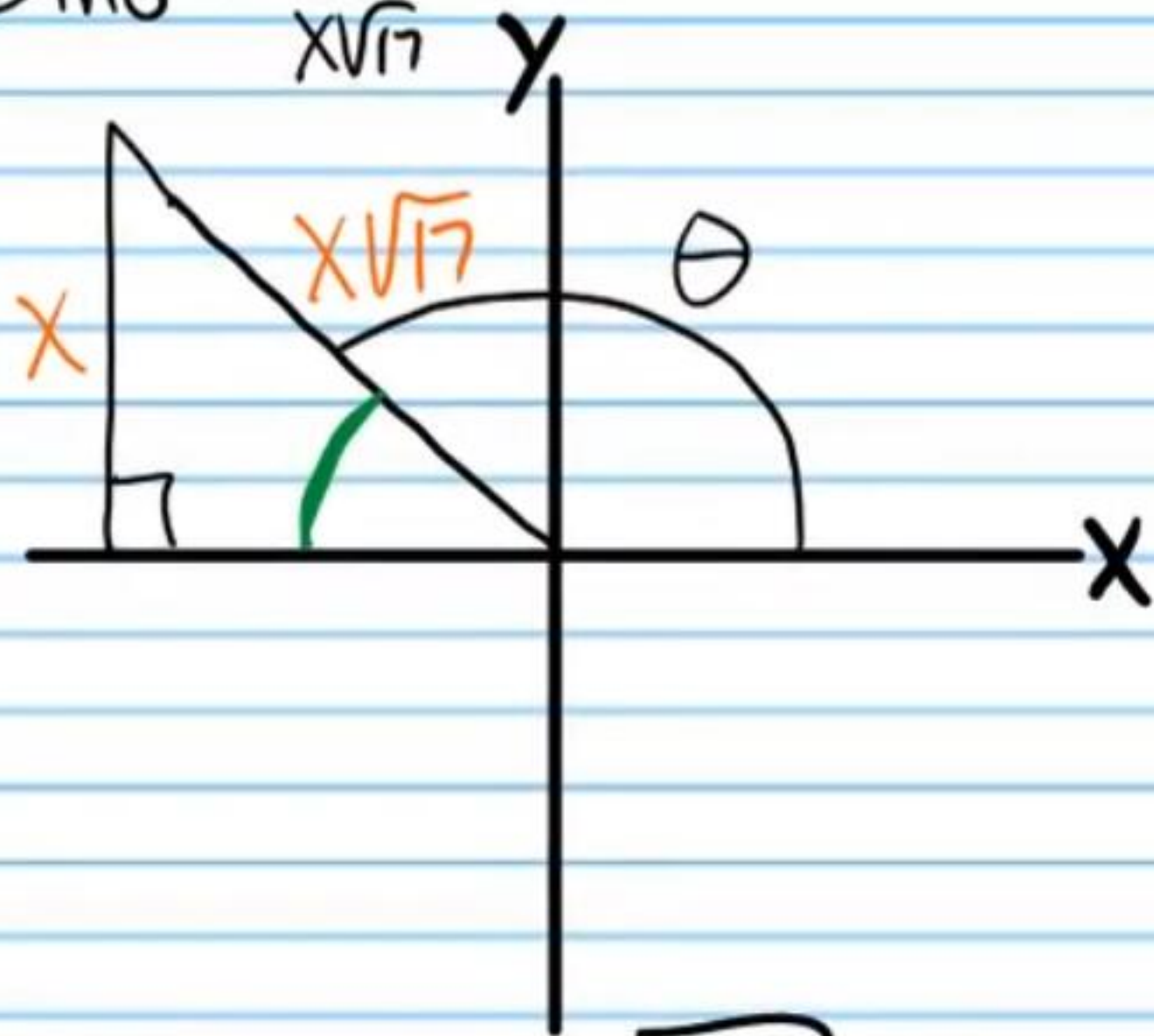
$$-0.342 = -0.342 = -0.342$$



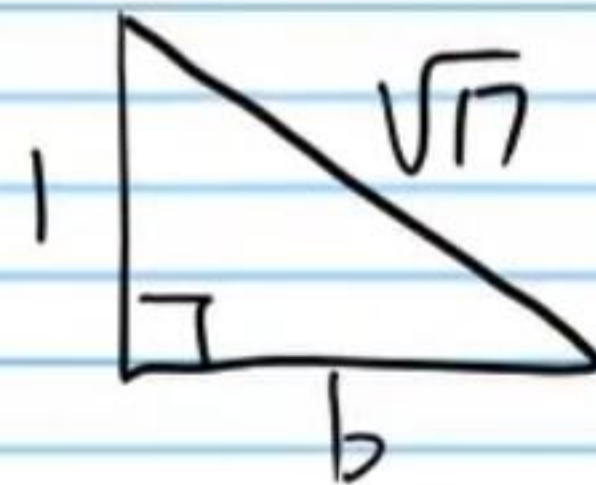
Finding the Five Remaining Trigonometric Ratios Given One

- ⑥ If $\sin \theta = \frac{1}{\sqrt{17}}$, and θ terminates in quadrant II, find the values of the five remaining basic trigonometric functions of θ .

$$\sin \theta = \frac{x}{x\sqrt{17}}$$



$$x=1$$



$$a^2 + b^2 = c^2$$

$$1^2 + b^2 = (\sqrt{17})^2$$

$$1 + b^2 = 17$$

$$b^2 = 16$$

$$b = \pm 4$$

$$\text{i) } \csc \theta = \frac{\sqrt{17}}{1} = \boxed{\sqrt{17}}$$

$$\text{ii) } \sec \theta = \boxed{\frac{\sqrt{17}}{-4}}$$

$$\text{iii) } \cot \theta = \frac{-4}{1} = \boxed{-4}$$

$$\text{iv) } \cos \theta = \frac{-4}{\sqrt{17}} = -\frac{4}{\sqrt{17}} \cdot \frac{\sqrt{17}}{\sqrt{17}} = \boxed{-\frac{4\sqrt{17}}{17}}$$

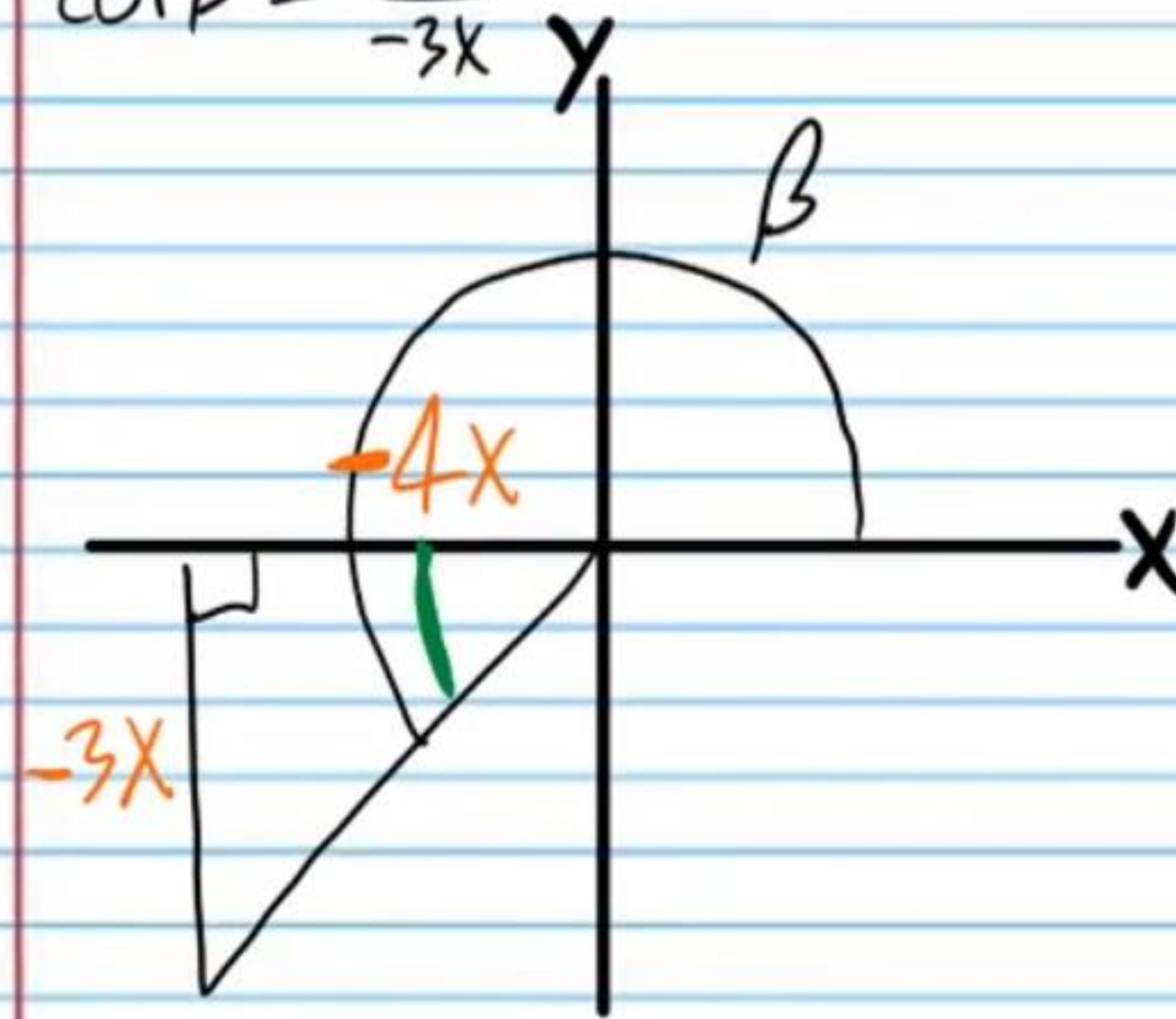
$$\text{v) } \tan \theta = \boxed{-\frac{1}{4}}$$



- ⑦ If $\cot \beta = \frac{4}{3}$, and β terminates in quadrant III, find the values of the five remaining basic trigonometric functions of β .

$$\cot \beta = \frac{-4x}{-3x}$$

$$x=1$$



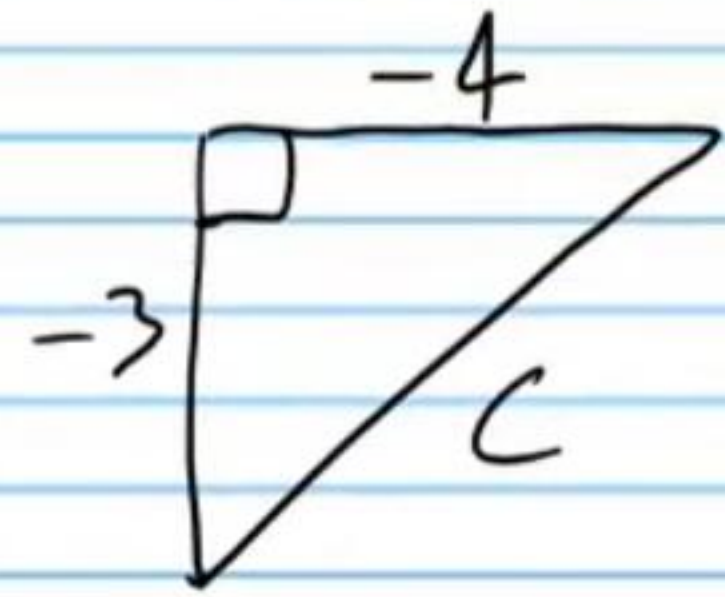
$$\text{i) } \cos \beta = \boxed{-\frac{4}{5}}$$

$$\text{ii) } \sin \beta = \boxed{-\frac{3}{5}}$$

$$\text{iii) } \csc \beta = \boxed{\frac{5}{-3}}$$

$$\text{iv) } \sec \beta = \boxed{\frac{5}{-4}}$$

$$\text{v) } \tan \beta = \frac{-3}{-4} = \boxed{\frac{3}{4}}$$



$$a^2 + b^2 = c^2$$

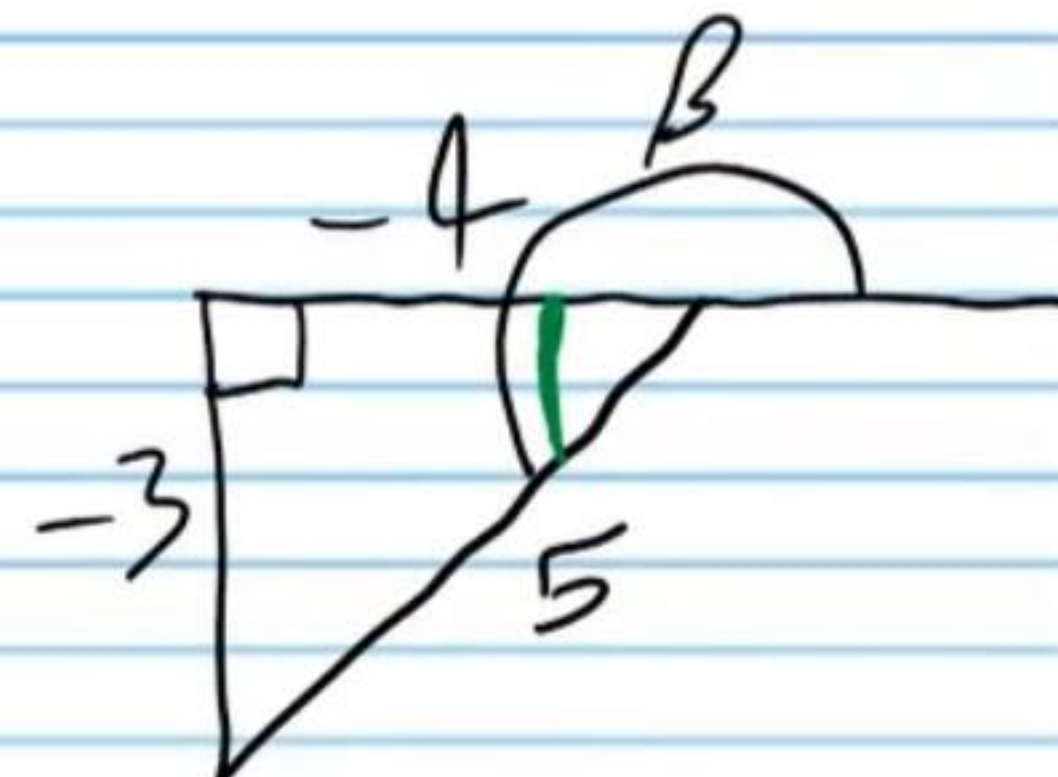
$$(-4)^2 + (-3)^2 = c^2$$

$$16 + 9 = c^2$$

$$25 = c^2$$

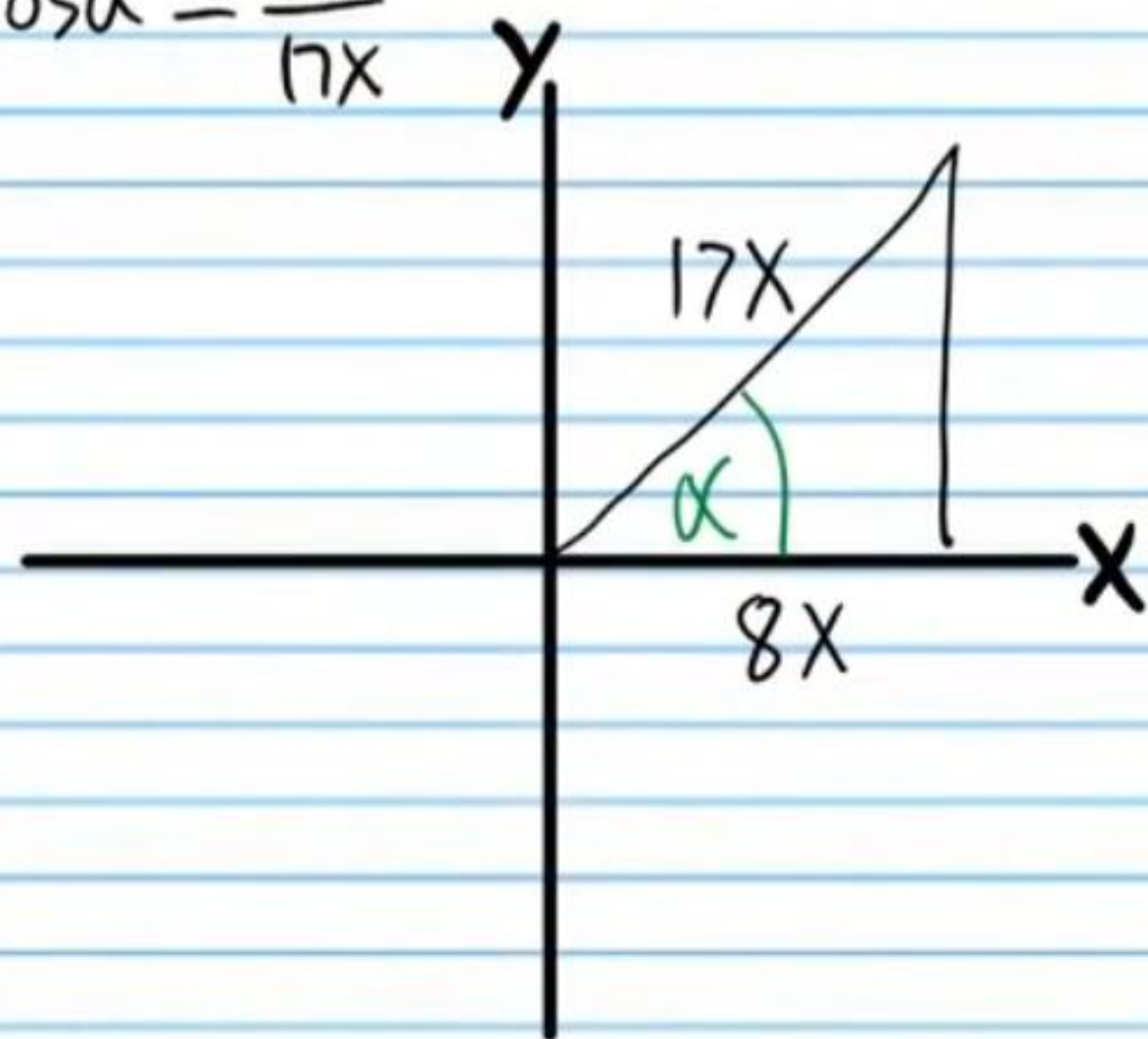
$$c = \pm 5$$

$$c = 5$$

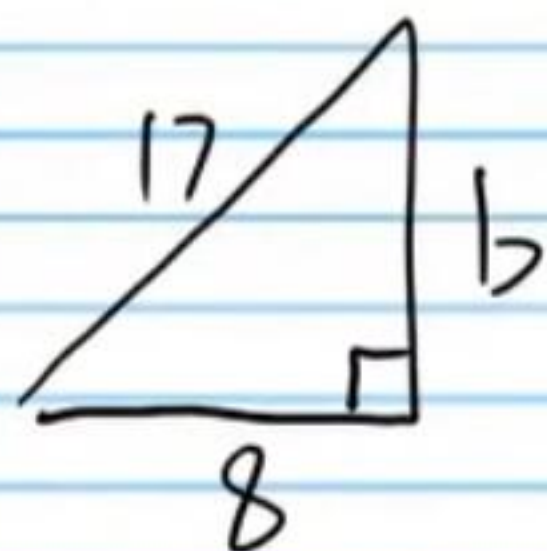


- 68) If $\cos \alpha = \frac{8}{17}$, and α terminates in quadrant I, find the values of the five remaining basic trigonometric functions of α .

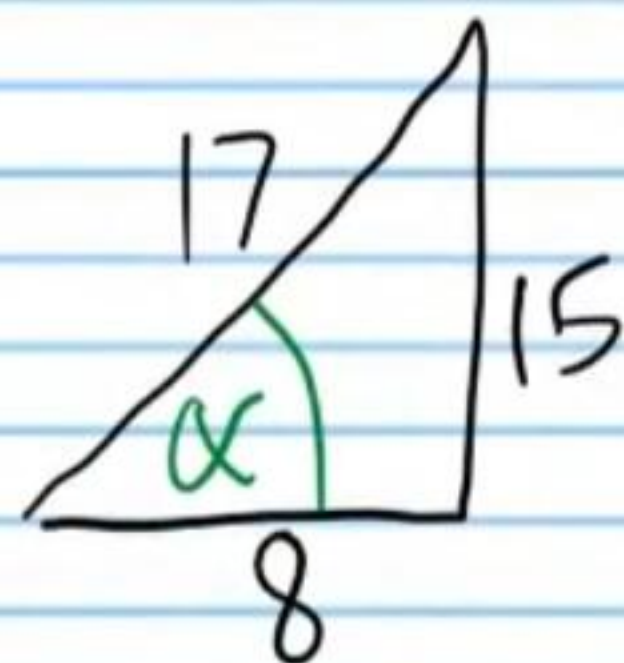
$$\cos \alpha = \frac{8x}{17x}$$



$$x=1$$



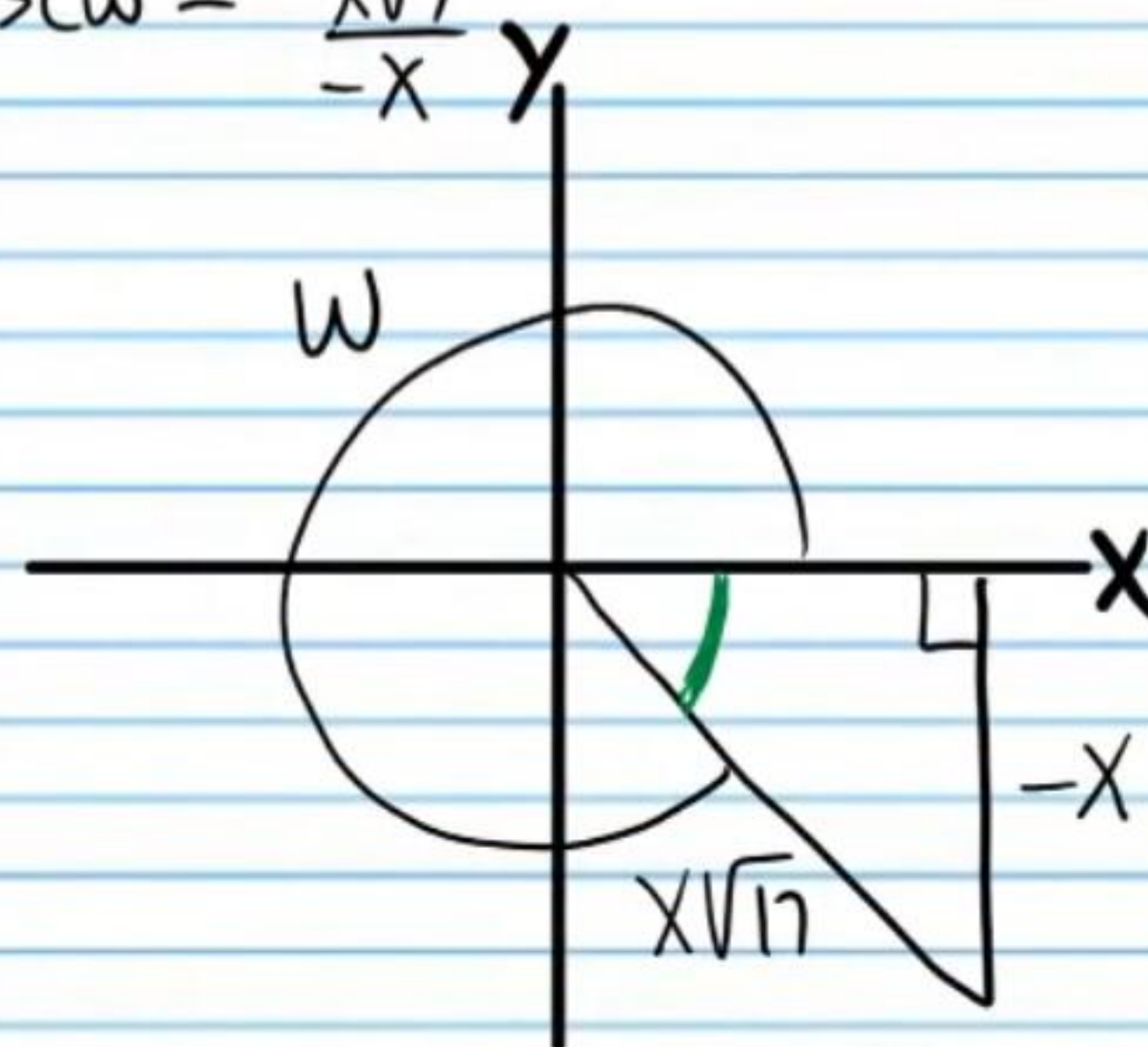
$$\begin{aligned} a^2 + b^2 &= c^2 \\ 8^2 + b^2 &= 17^2 \\ 64 + b^2 &= 289 \\ b^2 &= 225 \\ b &= \pm 15 \\ b &= 15 \end{aligned}$$



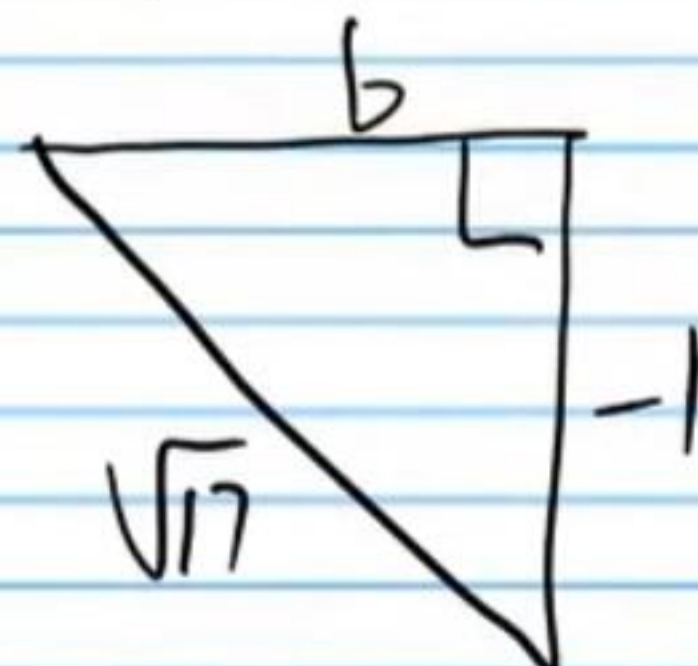
$$\begin{aligned} \text{i) } \sin \alpha &= \boxed{\frac{15}{17}} \\ \text{ii) } \tan \alpha &= \boxed{\frac{15}{8}} \\ \text{iii) } \csc \alpha &= \boxed{\frac{17}{15}} \\ \text{iv) } \sec \alpha &= \boxed{\frac{17}{8}} \\ \text{v) } \cot \alpha &= \boxed{\frac{8}{15}} \end{aligned}$$

- 69) If $\csc w = -\sqrt{17}$, and w terminates in quadrant IV, find the values of the five remaining basic trigonometric functions of w .

$$\csc w = \frac{x\sqrt{17}}{-x}$$

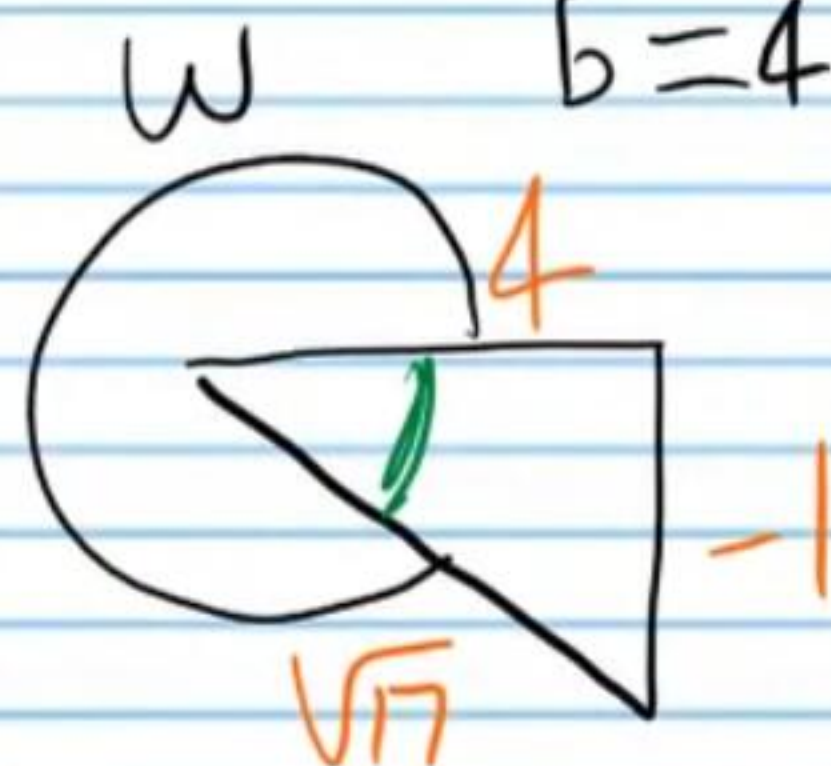


$$x=1$$



$$\begin{aligned} a^2 + b^2 &= c^2 \\ (-1)^2 + b^2 &= (\sqrt{17})^2 \\ 1 + b^2 &= 17 \end{aligned}$$

$$\begin{aligned} b^2 &= 16 \\ b &= \pm 4 \\ b &= 4 \end{aligned}$$

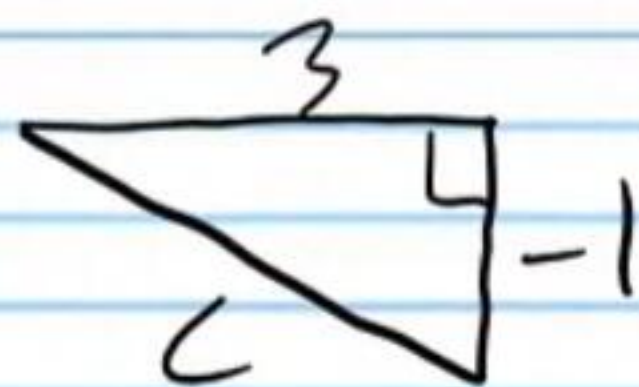
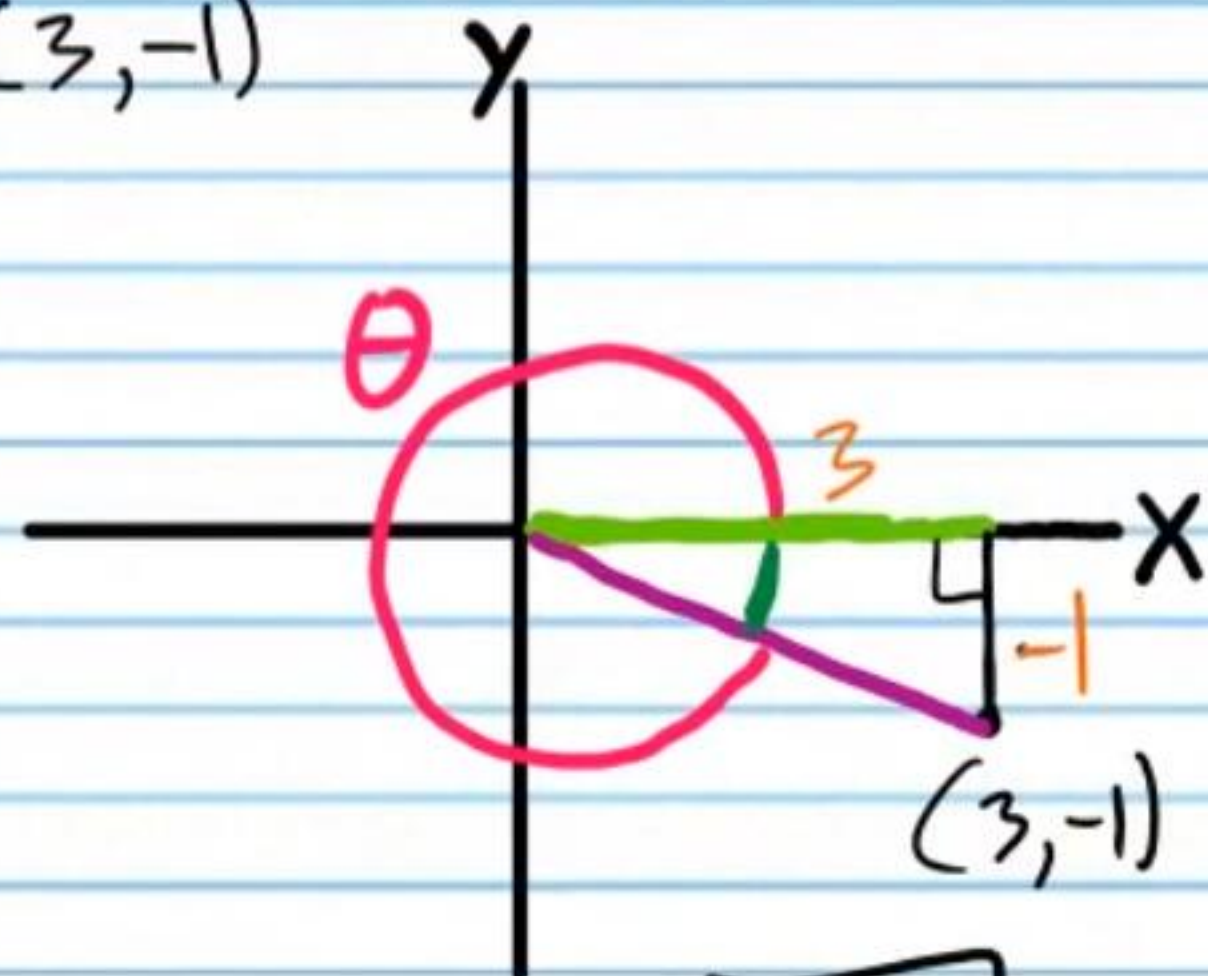


$$\begin{aligned} \text{i) } \cos w &= \frac{4}{\sqrt{17}} = \boxed{\frac{4\sqrt{17}}{17}} \\ \text{ii) } \sin w &= \frac{-1}{\sqrt{17}} = \boxed{-\frac{\sqrt{17}}{17}} \\ \text{iii) } \sec w &= \boxed{\frac{\sqrt{17}}{4}} \\ \text{iv) } \tan w &= \boxed{-\frac{1}{4}} \\ \text{v) } \cot w &= \frac{4}{-1} = \boxed{-4} \end{aligned}$$

Finding Trigonometric Values Given a Point on the Terminal Side of the Angle

The coordinates of points are given below. If each of these points lies on the terminal side of an angle in standard position, find the values of the six basic trigonometric functions of the angle.

• 70) $(3, -1)$



$$a^2 + b^2 = c^2$$

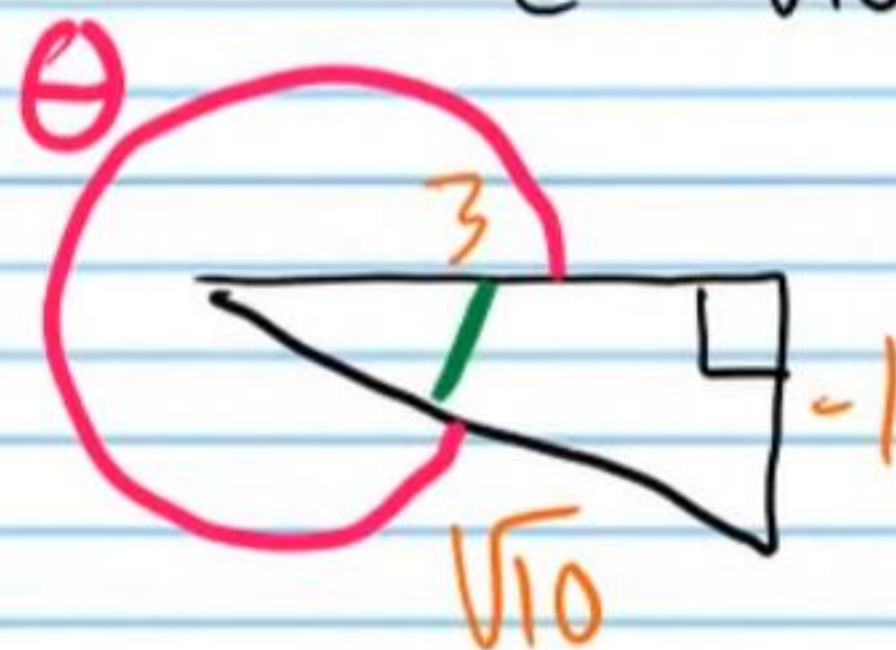
$$3^2 + (-1)^2 = c^2$$

$$9 + 1 = c^2$$

$$10 = c^2$$

$$c = \pm \sqrt{10}$$

$$c = \sqrt{10}$$



$$\text{i) } \sin \theta = \frac{-1}{\sqrt{10}} = \boxed{-\frac{\sqrt{10}}{10}}$$

$$\text{ii) } \cos \theta = \frac{3}{\sqrt{10}} = \boxed{\frac{3\sqrt{10}}{10}}$$

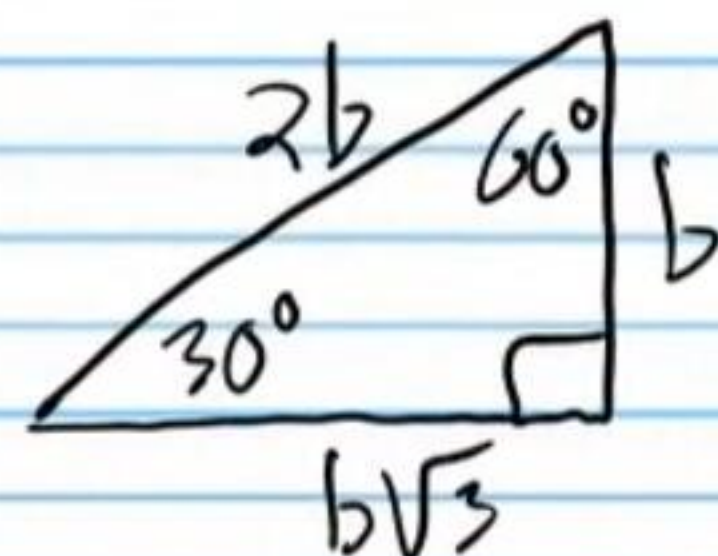
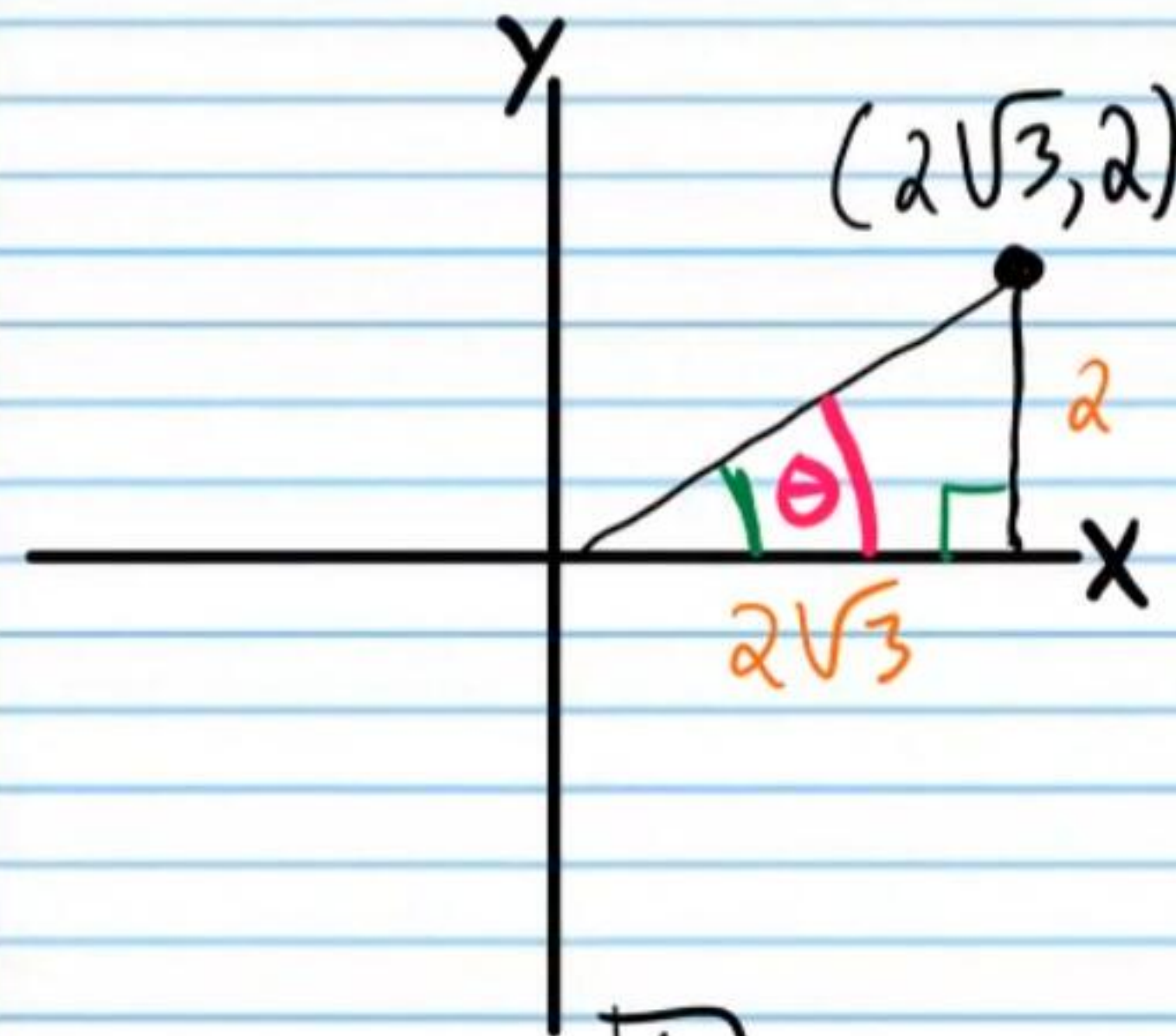
$$\text{iii) } \tan \theta = \boxed{-\frac{1}{3}}$$

$$\text{iv) } \csc \theta = \frac{\sqrt{10}}{-1} = \boxed{-\sqrt{10}}$$

$$\text{v) } \sec \theta = \boxed{\frac{\sqrt{10}}{3}}$$

$$\text{vi) } \cot \theta = \frac{3}{-1} = \boxed{-3}$$

71) $(2\sqrt{3}, 2)$



$$b = 2$$

$$2b = 2(2) = 4$$

$$\text{i) } \sin \theta = \frac{2}{4} = \boxed{\frac{1}{2}}$$

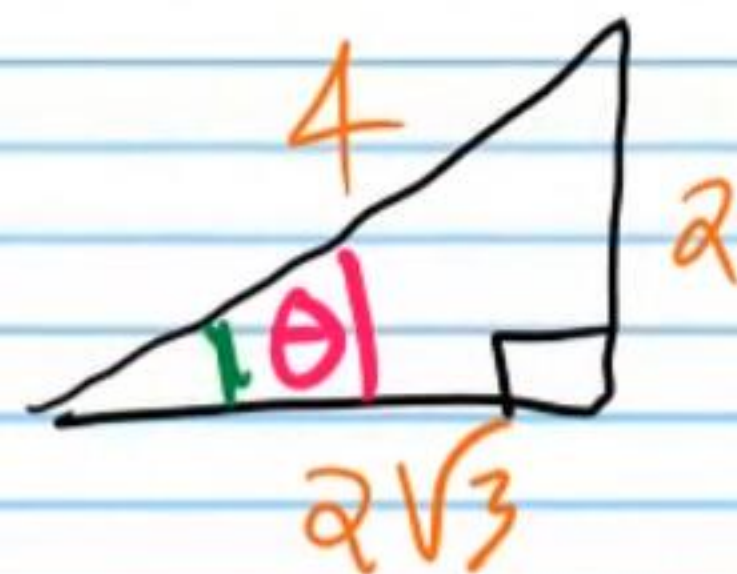
$$\text{ii) } \cos \theta = \frac{2\sqrt{3}}{4} = \boxed{\frac{\sqrt{3}}{2}}$$

$$\text{iii) } \tan \theta = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}} = \boxed{\frac{\sqrt{3}}{3}}$$

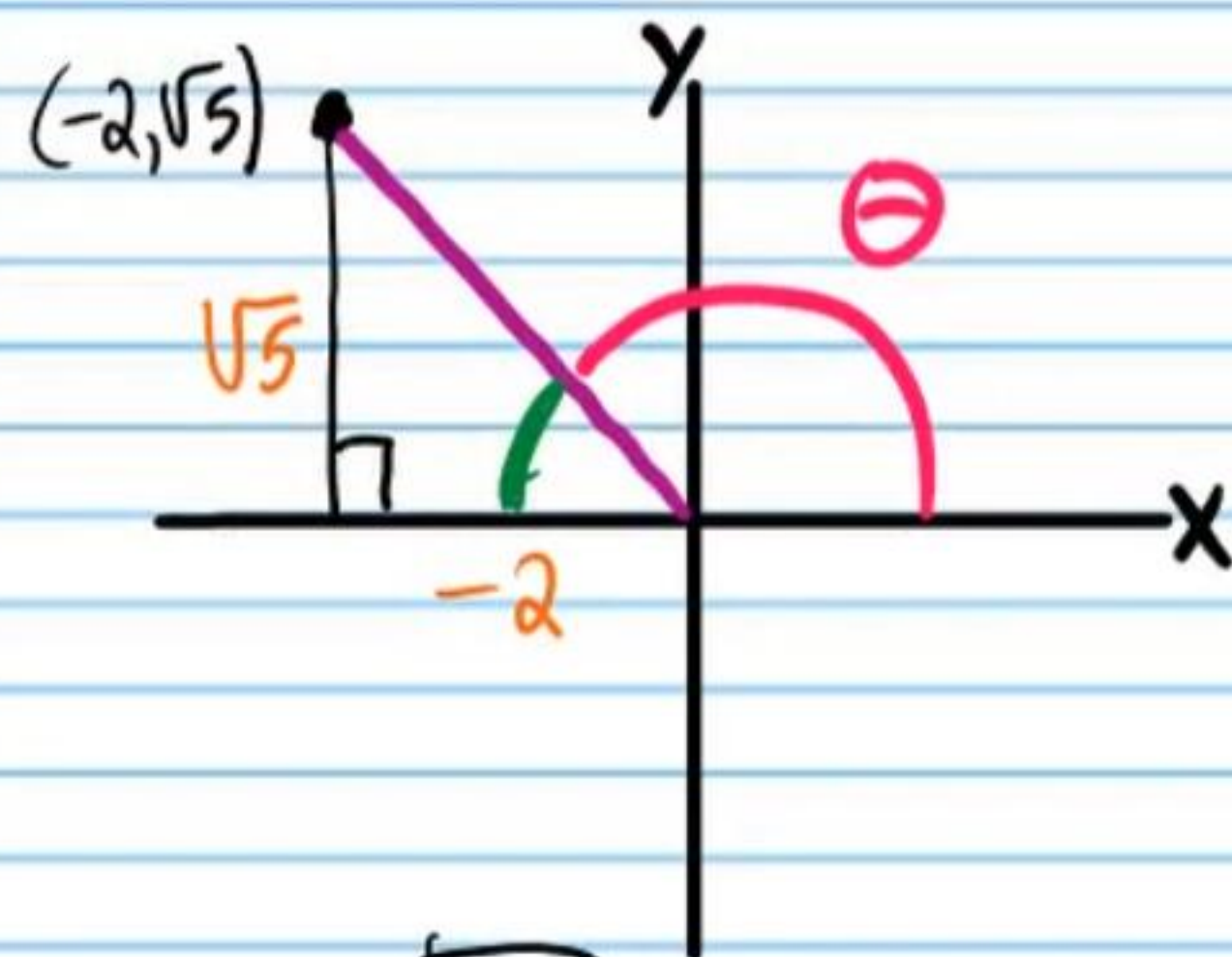
$$\text{iv) } \csc \theta = \frac{4}{2} = \boxed{2}$$

$$\text{v) } \sec \theta = \frac{4}{2\sqrt{3}} = \frac{2}{\sqrt{3}} = \boxed{\frac{2\sqrt{3}}{3}}$$

$$\text{vi) } \cot \theta = \frac{2\sqrt{3}}{2} = \boxed{\sqrt{3}}$$



72) $(-2, \sqrt{5})$



$$\text{i) } \sin \theta = \frac{\sqrt{5}}{3}$$

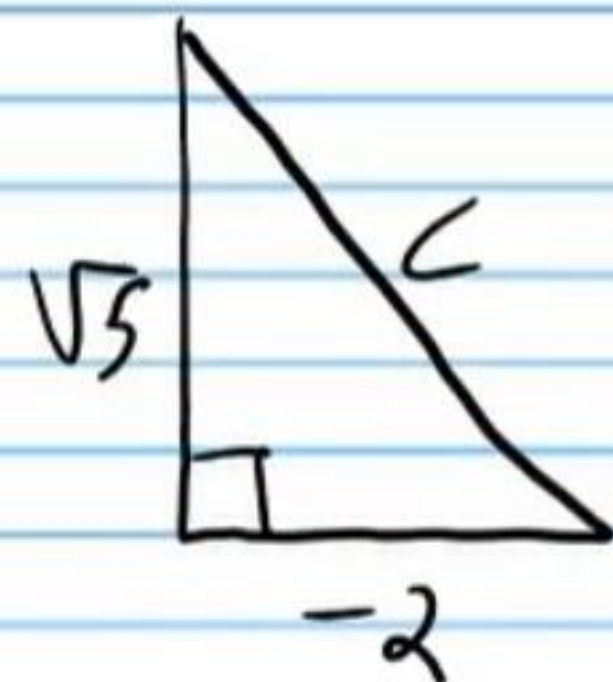
$$\text{ii) } \cos \theta = \frac{-2}{3}$$

$$\text{iii) } \tan \theta = \frac{\sqrt{5}}{-2}$$

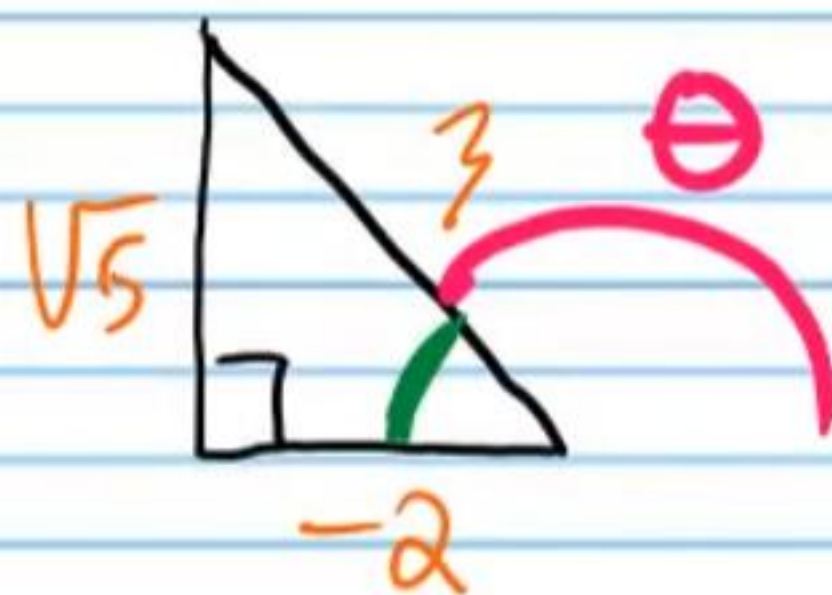
$$\text{iv) } \csc \theta = \frac{3}{\sqrt{5}} = \frac{3\sqrt{5}}{5}$$

$$\text{v) } \sec \theta = \frac{3}{-2}$$

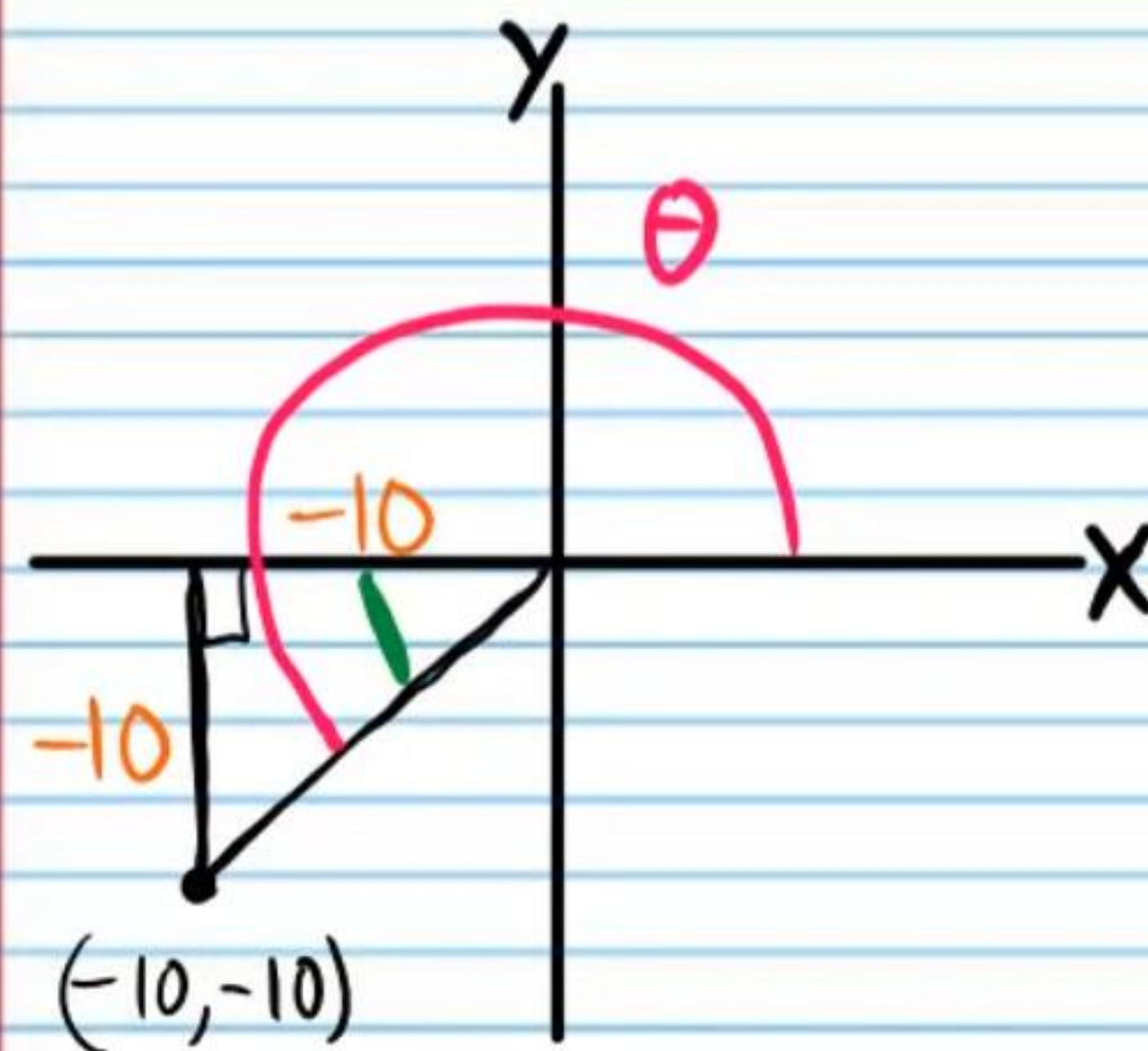
$$\text{vi) } \cot \theta = \frac{-2}{\sqrt{5}} = \frac{-2\sqrt{5}}{5}$$



$$\begin{aligned} a^2 + b^2 &= c^2 \\ (\sqrt{5})^2 + (-2)^2 &= c^2 \\ 5 + 4 &= c^2 \\ 9 &= c^2 \\ c &= \pm 3 \\ c &= 3 \end{aligned}$$



73) $(-10, -10)$



$$\text{i) } \sin \theta = \frac{-10}{10\sqrt{2}} = -\frac{1}{\sqrt{2}} = \frac{-\sqrt{2}}{2}$$

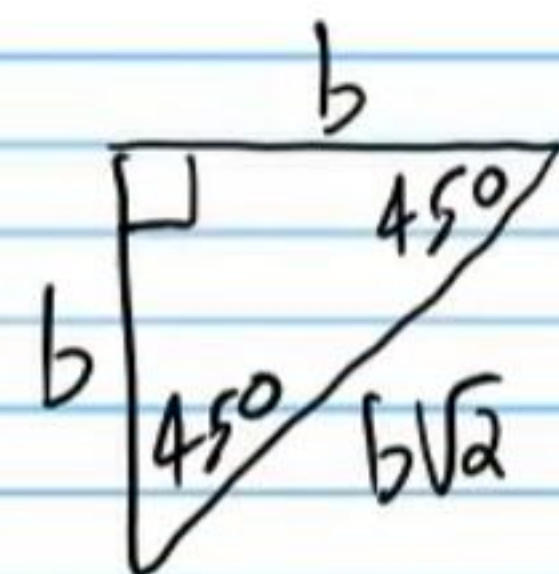
$$\text{ii) } \cos \theta = \frac{-10}{10\sqrt{2}} = \frac{-\sqrt{2}}{2}$$

$$\text{iii) } \tan \theta = \frac{-10}{-10} = 1$$

$$\text{iv) } \csc \theta = \frac{10\sqrt{2}}{-10} = -\sqrt{2}$$

$$\text{v) } \sec \theta = \frac{-10}{-10} = 1$$

$$\text{vi) } \cot \theta = \frac{-10}{-10} = 1$$



$$\begin{aligned} b &= -10 \\ b\sqrt{2} &= -10\sqrt{2} \\ &= 10\sqrt{2} \end{aligned}$$

